$\pi^2 - 18\log^2\varphi$ IS IRRATIONAL

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ABSTRACT. Let $\varphi = (1+\sqrt{5})/2$. We prove that $\pi^2 - 18\log^2\varphi$ is irrational. Equivalently, the value $\chi_2(\sqrt{5}-2) = \pi^2/24 - \frac{3}{4}\log^2\varphi$ of Legendre's chi function is irrational. The proof transposes our proof that $\chi_2(1/3)$ is irrational to the point $z = \sqrt{5}-2$, a unit of norm -1 in $\mathbb{Q}(\sqrt{5})$. We take poles at the even integers $\pm 2j$, $|j| \le h/2$, and atoms at the odd positive integers. A reflection $t \mapsto -t$, combined with the Galois conjugation $z \mapsto -1/z$, shows that the Hankel determinants are values of polynomials $\Delta_h(X) \in \mathbb{Q}[X]$, although their entries lie in $\mathbb{Q}(\sqrt{5})$; so the conjugate place, where the defining series diverges, costs nothing. Because z is a unit, there are no special primes. The odd primes cost at most $4/3$ per h^2, through a bordered Lagrange Gram identity and a class count. The prime 2 costs nothing at the scale h^2, because $1 - z^2 = 4z$. The size is bounded by a discrete log-gas argument against a dual certificate checked in interval arithmetic, which gives at most -1.3592 per h^2; nonvanishing comes from a reduction modulo a prime $\ell = (h+1)/3$. The margin is $-0.02588\ldots$ per h^2. The theorem is formalised in Lean 4 with Mathlib, using only Lean's three standard axioms; the formal proof of the analytic bound follows a different route from the one written here. We also prove a two-point statement at units of a family of biquadratic fields: for $n \in \{730, 740, 750, 760, 800, 1000\}$, either $\chi_2(z) + \chi_2(w) \notin \mathbb{Q}$ or $\chi_2(z) - \chi_2(w) \notin \sqrt{n^2-16}\,\mathbb{Q}$, where $z > w$ are the two roots in $(0,1)$ of $X^4 + nX^3 + 2X^2 - nX + 1$. This second result is computer-assisted and not formalised.

CONTENTS

1. Introduction 2
1.1. Context 3
1.2. The method 4
1.3. The formalisation 6
1.4. Two points 6
1.5. Organisation and notation 6
2. The construction 6
2.1. Data 6

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2.2. The pole expansion 7
2.3. The reflection 8
2.4. Valuations and the criterion 8
3. Denominators I: the odd primes 9
3.1. The atom expansion and the class count 9
3.2. The plateau: a bordered Lagrange Gram identity 10
3.3. The sum over the odd primes 12
4. Denominators II: the prime 2 12
5. The size of $\Delta_h(T)$: Theorem B′ 14
5.1. The equilibrium problem and weak duality 14
5.2. The theorem 14
5.3. The certificate 17
6. Nonvanishing: Lemma P′ 18
6.1. Setting 18
7. Proof of Theorem 1.1 19
8. Two points 20
8.1. The statement 20
8.2. What changes at two points 21
8.3. The certificates 23
8.4. Scope 24
9. The formalisation 25
9.1. What is formalised 25
9.2. The statement 25
9.3. Correspondence with the paper 26
9.4. Differences from the written proof 26
9.5. How the certificate is checked 27
9.6. Cost 28
9.7. What has to be trusted 28
10. Computations and reproducibility 28
10.1. Software 28
10.2. What each computation is used for 28
10.3. Numerical evidence 29
10.4. Controls 29
Acknowledgements and disclosure 30
References 30

1. Introduction

Let $\varphi = (1+\sqrt{5})/2$ be the golden ratio, and for $|q| \le 1$ let

$$\chi_2(q) = \sum_{n=0}^{\infty} \frac{q^{2n+1}}{(2n+1)^2} = \tfrac{1}{2}\big(\mathrm{Li}_2(q) - \mathrm{Li}_2(-q)\big)$$

be Legendre's chi function [17, §1.8].

Theorem 1.1 (Theorem U)**.** *The number*

$$\pi^2 - 18\log^2\varphi = 5.70143763069985955388\ldots$$

is irrational. Equivalently, the number

$$\chi_2(\sqrt{5}-2) = \frac{\pi^2}{24} - \frac{3}{4}\log^2\varphi = 0.23755990127916081474\ldots$$

is irrational.

The two forms are equivalent by the identity

$$\chi_2(\sqrt{5}-2) = \frac{\pi^2}{24} - \frac{3}{4}\log^2\varphi, \tag{1.1}$$

which follows from Landen's identity $\chi_2\bigl(\frac{1-x}{1+x}\bigr) + \chi_2(x) = \frac{\pi^2}{8} + \frac{1}{2}\log x\,\log\frac{1+x}{1-x}$ [17, App. A.2.1, (24)] at $x = \sqrt{5}-2 = \varphi^{-3}$, for which $(1-x)/(1+x) = \varphi^{-1}$, together with the classical value $\chi_2(\varphi^{-1}) = \pi^2/12 - \frac{3}{4}\log^2\varphi$. The formalisation proves (1.1) from these identities (Section 9).

1.1. **Context.** We have not found Theorem 1.1, or the irrationality of any number $\pi^2 + c\log^2\varphi$ with $c \in \mathbb{Q}\setminus\{0\}$, in the literature, and as far as we can see it does not follow from known results. By the analytic class number formula, $L(1,\chi_5) = 2\log\varphi/\sqrt{5}$ for the quadratic character χ_5 modulo 5, so the theorem can also be written as $6\zeta(2) - \frac{45}{2}L(1,\chi_5)^2 \notin \mathbb{Q}$.

- *The golden-ratio dilogarithm values are different numbers.* The classical evaluations

  $$\operatorname{Li}_2(\varphi^{-2}) = \tfrac{\pi^2}{15} - \log^2\varphi, \qquad \operatorname{Li}_2(\varphi^{-1}) = \tfrac{\pi^2}{10} - \log^2\varphi,$$
  $$\operatorname{Li}_2(-\varphi^{-1}) = -\tfrac{\pi^2}{15} + \tfrac{1}{2}\log^2\varphi, \qquad \operatorname{Li}_2(-\varphi) = -\tfrac{\pi^2}{10} - \log^2\varphi$$

  (Landen; see [17, 18]) all lie in $\mathbb{Q}\pi^2 + \mathbb{Q}\log^2\varphi$. Normalised to the form $\pi^2 + c\log^2\varphi$ they have $c = -15$, -10, $-\frac{15}{2}$ and $+10$, while Theorem 1.1 has $c = -18$. Since $\log\varphi$ is transcendental, $\pi^2 + c\log^2\varphi$ can be algebraic for at most one $c \in \overline{\mathbb{Q}}$; this does not say which. The irrationality of one member of the family says nothing about another, and as far as we know the irrationality of each of the four classical values is open.
- *Homogeneous statements.* The theorem of Gel'fond and Schneider [1, Ch. 2], applied to $e^{\log\varphi} = \varphi$ and $e^{i\pi} = -1$, shows that $\log\varphi/(i\pi)$ is transcendental, so $\log^2\varphi/\pi^2 \notin \overline{\mathbb{Q}}$. This says that π^2 and $\log^2\varphi$ are linearly independent over $\mathbb{Q}$; it says nothing about a relation that also involves 1. The only member of the family $\pi^2 + c\log^2\varphi$, $c \in \mathbb{Q}$, whose irrationality we know to be established is $c = 0$. The linear independence of $1, \pi^2, \log^2\varphi$ over $\mathbb{Q}$ is open as far as we know.
- *Criteria at algebraic points.* The linear independence criteria for polylogarithms and Lerch functions at algebraic points [3, 5, 4, 12], and the explicit form of the Bombieri–André principle of global relations [20], need a point that is small in comparison with its

height. Here $\chi_2(\sqrt{5}-2) = \frac{1}{2}(\mathrm{Li}_2(z) - \mathrm{Li}_2(-z)) = z + \frac{z}{4}\Phi_2(\frac{1}{2}, z^2)$, where $\Phi_2(x,q) = \sum_{k\ge 0} q^{k+1}/(k+x+1)^2$ is the Lerch function and $z = \sqrt{5}-2 = \varphi^{-3}$ is a unit of height $\frac{3}{2}\log\varphi = 0.7218\ldots$ whose conjugate $-\varphi^3$ has absolute value > 1, so that $|z|\,|\sigma z| = 1$. Evaluating the criteria in the form they are stated, their key quantity is negative by a wide margin at this point: about -26.1 for [12, Thm. 2.2], and about -15.1 and -14.5 for the two ways of applying [4, Thm. 1] (through $\mathrm{Li}_2(\pm z)$, or through $\Phi_2(\frac{1}{2}, z^2)$). The Bombieri–André bound of [20], applied to this system, needs by our estimate a height of about 1450. These evaluations are ours and have not been checked by the authors of the criteria.

- *Rational points.* The irrationality and linear independence results for dilogarithms at rational points [8, 9, 22, 23, 30], the recent claims [21, 25] (unrefereed), and our result $\chi_2(1/3), \chi_2(1/5) \notin \mathbb{Q}$ [14] all concern rational arguments, as do the weight-two results of Calegari, Dimitrov and Tang [2] (at sixth roots of unity, and for products $\log(1+1/m)\log(1+1/n)$) and Hata's irrationality of $\log(1+1/q)\log(1-1/q)$ [10]. The results on products of logarithms are homogeneous and contain no π^2.
- *The closest analogue.* The nearest result we know is $\mathrm{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2}\log^2 2 \notin \mathbb{Q}$, proved in Lean by Q. Tang [26] and, with many more rational points, by S. Liu [19]; neither is accompanied by a refereed paper. It has the same shape, $\pi^2 + c\log^2$ of a number, with $c = -6$ and the number 2 instead of a unit. Over $\mathbb{Q}(\sqrt{5})$ we have found nothing comparable.

Our searches for prior work, in four passes, covered the web, arXiv (through its API), zbMATH Open, OpenAlex, Crossref, Semantic Scholar (only partly, because of rate limits), the Hexagon repository, the `openai/math` manuscript list and recent Lean repositories (the last pass on 7 November 2026). They did not cover MathSciNet or Google Scholar, and no expert has been consulted. The searches were carried out by AI systems.

1.2. **The method.** Put

$$z := \sqrt{5}-2 = \varphi^{-3}, \qquad y := z^2, \qquad T := \chi_2(z) = \sum_{t\ \mathrm{odd}\ge 1} \frac{z^t}{t^2}.$$

Then $z^2 + 4z - 1 = 0$, so $1 - y = 4z$, and the Galois conjugate of z is $-1/z$: z is a unit of norm -1 in $\mathbb{K} := \mathbb{Q}(\sqrt{5})$. The proof transposes the proof of [14] for $\chi_2(1/q)$ to this point. For even h we take the $K := h+1$ poles $x_m := 2(m - h/2)$, i.e. the even integers in $[-h, h]$, put $D(t) := \prod_m (t - x_m)$, and consider the functional

$$\Lambda(g) := \sum_{t\ \mathrm{odd}\ge 1} z^t \Big(\frac{g}{D}\Big)'(t) \qquad (g \in \mathbb{Q}[t]).$$

Partial fractions give $\Lambda = A + T\,B$ with $\mathbb{K}$-valued functionals A, B, so the Hankel determinant $\det(\Lambda(t^{i+j}))_{i,j<h}$ is the value at $X = T$ of $\Delta_h(X) := \det(A(t^{i+j}) + XB(t^{i+j}))_{i,j<h}$, a polynomial of degree at most h.

What differs from [14] is the *reflection* (Lemma 2.2). The pole set is symmetric under $t \mapsto -t$, $D(-t) = -D(t)$, and the conjugation $\sigma\colon z \mapsto -1/z$ maps the partial sums of χ_2 at the poles x to those at $-x$. Together these give $\sigma(\Lambda_X(t^m)) = (-1)^m\Lambda_X(t^m)$, which forces $\Delta_h \in \mathbb{Q}[X]$ although its entries lie in $\mathbb{K}$. So we never pay for the conjugate place, where $|\sigma z| = \varphi^3 > 1$ and the defining series diverges. The three inputs are then as in [14].

(1) *Denominators* (Sections 3–4). Because z is a unit, the moments of Λ only have the denominators $(1-y)^{b+1} = (4z)^{b+1}$, so there are no special primes, and no gain prime either. Every odd prime, including 5, satisfies the same bounds. Primes $p < K/3$ obey a plateau bound $2h\,\mathrm{lev}^*(p) + v_p(h!)$, which we prove through a bordered version of the Lagrange Gram identity (Lemmas 3.4 and 3.5), and primes $K/3 \le p < K$ obey a class count $3(K-p)$ (Lemmas 3.2 and 3.3). The prime number theorem sums these to $\frac{2}{3} + \frac{2}{3} = \frac{4}{3}$ per h^2 (Theorem 3.7). The prime 2 costs only $O(h\log h)$ (Lemma 4.1), because $v_2(1-y) = 2$.

(2) *Size* (Section 5). A discrete log-gas argument bounds $h^{-2}\log|\Delta_h(T)|$ by the value Φ of any dual certificate for a constrained equilibrium problem with external field $x\log z - \frac{1}{2}\int_{x-1}^{x+1}\log|s|\,ds$ (Theorem 5.1). An explicit certificate, checked in interval arithmetic, gives $\Phi \le -1.3592161$ (Theorem 5.3).

(3) *Nonvanishing* (Section 6). For $h = 3\ell - 1$ with $\ell \ge 7$ prime, the Gram matrix of Λ in the Lagrange basis is block diagonal modulo a prime of $\mathbb{K}$ above ℓ, with 3×3 blocks in closed form, so $\Delta_h(a/b) \ne 0$ whenever $\ell^2 \nmid b$ (Lemma 6.1). Unlike in [14], every prime $\ell \ge 7$ works.

Table 1 collects the contributions.

contribution (per h^2, lim sup)	source	value
odd primes, including 5	Thm. 3.7	$4/3 = 1.3333333$
$p = 2$	Lemma 4.1	0
special primes	(z is a unit)	none
arithmetic		1.3333333
analytic Φ	Thms. 5.1, 5.3	≤ -1.3592161
margin		≤ -0.025882

TABLE 1. The exponential budget. The analytic value is the upper endpoint -1.3592161834586374 of the certified Arb ball, rounded up; the margin is $-0.0258828501\ldots$, rounded up.

The margin is about 1.9% of the arithmetic part. It is smaller than in [14] for $q = 3$ in relative terms, but no constant in the budget is estimated: each is a proved upper bound.

1.3. **The formalisation.** Theorem 1.1 and the identity (1.1) are formalised in Lean 4 [6] with Mathlib [28] and checked by the Lean kernel. The headline theorems are `Chi2U.chi2_sqrt5_sub_two_irrational` ($\chi_2(\sqrt{5}-2) \notin \mathbb{Q}$), `Chi2U.golden_irrational'` ($\pi^2 - 18\log^2\varphi \notin \mathbb{Q}$) and `Chi2U.chi2_sqrt5_sub_two_eq` (the identity (1.1)). They depend only on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`); there is no `sorry` and no `native_decide`. The formal proof of the analytic bound follows the lattice-energy route of [14], not the route of Section 5, and uses its own, coarser certificate; Section 9 describes both differences.

1.4. **Two points.** Section 8 transposes the proof once more, to units of the biquadratic fields $\mathbb{Q}(z)$ where z is a root of $X^4+nX^3+2X^2-nX+1$. There the reflection removes the two conjugates $-1/z$ and $-1/w$, where the series diverge, and the remaining conjugate w is a second convergent place, which enters through a norm. The result is a disjunction, proved separately for six values of n, with interval-arithmetic certificates. The smallest instance is the following.

Theorem 1.2 (M1 at $n = 730$)**.** *Let $z > w$ be the two roots in $(0,1)$ of $X^4 + 730X^3 + 2X^2 - 730X + 1$. Then at least one of the following holds:*

$$\frac{\pi^2}{8} - \frac{1}{8}\log^2\bigl(366+\sqrt{133955}\bigr) + \frac{1}{8}\log^2\bigl(364+11\sqrt{1095}\bigr) \notin \mathbb{Q},$$

$$\chi_2(z) - \chi_2(w) \notin \sqrt{1101}\,\mathbb{Q}.$$

The first number is $\chi_2(z) + \chi_2(w)$, by Landen's identity. Theorem 1.2 is not formalised.

1.5. **Organisation and notation.** Section 2 sets up the functional, proves the pole expansion and the reflection, and states the criterion. Sections 3 and 4 bound the denominators, Section 5 bounds the size, Section 6 proves nonvanishing, and Section 7 assembles the proof. Section 8 treats two points, Section 9 describes the formalisation, and Section 10 lists the computations.

Throughout, p and ℓ denote primes, v_p is the p-adic valuation, $\lfloor\cdot\rfloor$ is the floor, $(x)_+ = \max(x,0)$, and $[\,P\,]$ is 1 if P holds and 0 otherwise. For a function G on $\mathbb{Z}_{\geq 0}$, $\Delta G(s) = G(s+1) - G(s)$ is the forward difference. We write $\theta(x) = \sum_{p\leq x}\log p$ and $\psi(x) = \sum_{p^e\leq x}\log p$, and $s_2(b)$ for the binary digit sum of b, so $v_2(b!) = b - s_2(b)$.

2. The construction

2.1. **Data.** Let $\mathbb{K} = \mathbb{Q}(\sqrt{5})$, embedded in $\mathbb{R}$ with $\sqrt{5} > 0$, and let σ be its nontrivial automorphism. With $z = \sqrt{5} - 2$ and $y = z^2$ we have

$$z^2 + 4z - 1 = 0, \qquad 1 - y = 4z, \qquad \sigma(z) = -1/z.$$

So z, y and $1-y$ are units at every odd prime, including 5, and $1-y$ has 2-adic valuation 2.

Fix an even $h \geq 2$ and put $J := h/2$ and $K := h+1$. The poles are $x_m := 2(m-J)$ for $0 \leq m < K$, i.e. the even integers $\pm 2j$, $|j| \leq J$, and

$$D(t) := \prod_{m<K} (t - x_m),$$

$$D'(x_m) = \prod_{l \neq m} 2(m-l) = (-1)^{K-1-m} 2^{K-1} m!\,(K-1-m)!.$$

For a rational function f regular at the odd positive integers and of polynomial growth put

$$\mathcal{M}(f) := \sum_{t \text{ odd} \geq 1} z^t f'(t),$$

an absolutely convergent series, and $\Lambda(g) := \mathcal{M}(g/D)$ for $g \in \mathbb{Q}[t]$. For an even integer x define the partial sums

$$P(x) := \begin{cases} \displaystyle\sum_{n \text{ odd},\, 0<n<|x|} \frac{z^n}{n^2}, & x < 0, \\ \displaystyle -\sum_{n \text{ odd},\, -x<n<0} \frac{z^n}{n^2}, & x > 0, \\ 0, & x = 0. \end{cases}$$

Only odd n with $|n| \leq |x| - 1 \leq h - 1$ occur.

2.2. **The pole expansion.**

Lemma 2.1 (Lemma 1′, pole expansion)**.** *For every even integer x, $\mathcal{M}\big(1/(t-x)\big) = -z^x\big(T - P(x)\big)$. For a polynomial F,*

$$\mathcal{M}(F) = z \sum_{b \geq 0} \Delta^b G(0)\, \frac{y^b}{(1-y)^{b+1}}, \qquad G(s) := F'(2s+1), \tag{2.1}$$

a finite sum. Consequently, writing $g/D = R_g + \sum_m g(x_m)/(D'(x_m)(t-x_m))$ with $R_g \in \mathbb{Q}[t]$,

$$\Lambda(g) = A(g) + T\,B(g),$$

$$A(g) := \mathcal{M}(R_g) + \sum_m \frac{g(x_m)}{D'(x_m)} z^{x_m} P(x_m), \qquad B(g) := -\sum_m \frac{g(x_m)}{D'(x_m)} z^{x_m},$$

with $A, B\colon \mathbb{Q}[t] \to \mathbb{K}$ linear.

Proof. $\mathcal{M}(1/(t-x)) = -\sum_{t \text{ odd} \geq 1} z^t/(t-x)^2$. With $n = t - x$ (odd, $n \geq 1 - x$) and $z^t = z^x z^n$, the sum is $z^x \sum_{n \text{ odd} \geq 1-x} z^n/n^2 = z^x(T - P(x))$ by the definition of P. For a polynomial, substitute $t = 2s+1$, expand $G(s) = \sum_b \Delta^b G(0) \binom{s}{b}$, and use $\sum_{s \geq 0} \binom{s}{b} y^s = y^b/(1-y)^{b+1}$. All series converge absolutely since $0 < z < 1$. □

For an indeterminate X put $\Lambda_X := A + X\,B$, $E_m := \Lambda_X(t^m) \in \mathbb{K}[X]$ and

$$\Delta_h(X) := \det\big(E_{i+j}\big)_{0\le i,j<h} \in \mathbb{K}[X], \qquad \deg \Delta_h \le h,$$
$$\Delta_h(T) = \det\big(\Lambda(t^{i+j})\big)_{i,j<h}.$$

We use the same notation for every finite family: if $a_0,\dots,a_{h-1}$ and $b_0,\dots,b_{h-1}$ have degrees $< h$ and coefficient matrices C_a, C_b, then $\det(\Lambda_X(a_i b_j)) = \det C_a \det C_b\, \Delta_h(X)$.

2.3. **The reflection.** Extend A and B to rational functions $f = R + \sum_m c_m/(t-x_m)$ ($R \in \mathbb{Q}[t]$, $c_m \in \mathbb{Q}$) by $A(f) := \mathcal{M}(R) + \sum_m c_m z^{x_m} P(x_m)$ and $B(f) := -\sum_m c_m z^{x_m}$, so that $\mathcal{M}(f) = A(f) + TB(f)$, and put $\tilde f(t) := f(-t)$.

Lemma 2.2 (Lemma R, reflection)**.** *For every such f, $\sigma(A(f)) + \sigma(B(f))X = -\big(A(\tilde f) + B(\tilde f)X\big)$.*

Proof. We have $\tilde f = \tilde R + \sum_m (-c_m)/(t-(-x_m))$, and the pole set is symmetric. *The X-part.* Since x_m is even, $\sigma(z^{x_m}) = (-1/z)^{x_m} = z^{-x_m}$, so $\sigma(B(f)) = -\sum_m c_m z^{-x_m} = -B(\tilde f)$. *The pole part of A.* For $x<0$, $\sigma(P(x)) = \sum_{0<n<|x|}(-1)^n z^{-n}/n^2 = -\sum_{-|x|<n'<0} z^{n'}/n'^2 = P(-x)$ (odd n, $n' = -n$), and likewise for $x>0$. Hence $\sigma\big(\sum_m c_m z^{x_m}P(x_m)\big) = \sum_m c_m z^{-x_m}P(-x_m)$, which is minus the pole part of $A(\tilde f)$. *The polynomial part.* For a polynomial R, $\mathcal{M}(R) = M_R(z)$ with $M_R(q) := \sum_{t\ \mathrm{odd}\ge 1} R'(t)q^t$, a rational function of q with rational coefficients and poles only at $q=\pm1$; so $\sigma(\mathcal{M}(R)) = M_R(-1/z)$. Since R' is a polynomial, the reciprocity of rational generating functions gives $\sum_{t\ \mathrm{odd}\ge 1} R'(t)q^t = -\sum_{t\ \mathrm{odd}\le -1} R'(t)q^t$ as rational functions, the right side being expanded at $q=\infty$: for $R'(t)=t^k$ apply $(q\,d/dq)^k$ to $\sum_{t\ \mathrm{odd}\ge1} q^t = q/(1-q^2) = -\sum_{t\ \mathrm{odd}\le -1} q^t$. At $q=-1/z$ the right side is $-\sum_{u\ \mathrm{odd}\ge1} R'(-u)(-z)^u = \sum_{u\ \mathrm{odd}\ge1} \tilde R'(-u)z^u = -M_{\tilde R}(z)$, because $\tilde R'(u) = -R'(-u)$. So $\sigma(\mathcal{M}(R)) = -\mathcal{M}(\tilde R)$. □

Corollary 2.3 (Corollary R)**.** *$\sigma(E_m) = (-1)^m E_m$ coefficientwise, and $\Delta_h(X) \in \mathbb{Q}[X]$.*

Proof. For $f = t^m/D$ we have $D(-t) = (-1)^K D(t) = -D(t)$ (K is odd), so $\tilde f = (-1)^{m+1} f$, and Lemma 2.2 gives $\sigma(E_m) = (-1)^m E_m$. Thus $E_m \in \mathbb{Q}[X]$ for even m and $E_m \in \sqrt{5}\,\mathbb{Q}[X]$ for odd m. The matrix $\tilde E_{ij} := E_{i+j}\sqrt{5}^{[i\ \mathrm{odd}]-[j\ \mathrm{odd}]}$ has entries in $\mathbb{Q}[X]$, and $\tilde E = \mathrm{diag}(\sqrt{5}^{[i\ \mathrm{odd}]})\,(E_{i+j})\,\mathrm{diag}(\sqrt{5}^{-[j\ \mathrm{odd}]})$, so $\det\tilde E = \Delta_h$. □

2.4. **Valuations and the criterion.** For a prime p and a prime $\mathfrak{p} \mid p$ of $\mathbb{K}$ let $v_\mathfrak{p}$ be normalised by $v_\mathfrak{p}(p) = 1$ (so $v(\sqrt{5}) = \frac12$ at $p=5$), extended to $\mathbb{K}[X]$ as the Gauss valuation (the minimum over the coefficients). For $p \ne 2, 5$ we write v_p; since 2 is inert and 5 ramified in $\mathbb{K}$ there is one prime above each

$p \in \{2,5\}$, and on $\mathbb{Q}[X]$ every $v_{\mathfrak{p}}$ restricts to v_p. For $\Delta_h \not\equiv 0$ put

$$n_p := -v_p(\Delta_h), \qquad F_h := \Delta_h \prod_p p^{n_p} \in \mathbb{Z}[X] \text{ (primitive)},$$

$$\log|F_h(T)| = \log|\Delta_h(T)| + \sum_p n_p \log p.$$

Upper bounds for n_p may be computed from the $\mathbb{K}$-valued entries at $\mathfrak{p} \mid p$, since $\Delta_h \in \mathbb{Q}[X]$.

Proposition 2.4 (Lemma E)**.** *Let $\xi \in \mathbb{R}$ and let $\mathcal{H}$ be an infinite set of positive integers. Suppose that for $h \in \mathcal{H}$ there are $F_h \in \mathbb{Z}[X]$ with $\deg F_h \le h$ and $\limsup_{h\in\mathcal{H}} h^{-2}\log|F_h(\xi)| < 0$, and that $F_h(a/b) \neq 0$ for all $h \in \mathcal{H}$, where $a \in \mathbb{Z}$, $b \ge 1$. Then $\xi \neq a/b$.*

Proof. If $\xi = a/b$, then $b^h F_h(a/b)$ is a nonzero integer, so $h\log b + \log|F_h(\xi)| \ge 0$ for $h \in \mathcal{H}$. Dividing by h^2 contradicts the hypothesis. □

This is the criterion of [14, Prop. 3.1], with the nonvanishing set allowed to depend on b.

3. Denominators I: the odd primes

Throughout this section p is an odd prime and $v = v_{\mathfrak{p}}$ for a prime $\mathfrak{p} \mid p$ of $\mathbb{K}$. Since D is monic with integer coefficients, $R_g \in \mathbb{Z}[t]$ for $g \in \mathbb{Z}[t]$, and by (2.1), with y and $1-y = 4z$ units at $\mathfrak{p}$,

$$v\big(\mathcal{M}(F)\big) \ge \min_{t \text{ odd} \ge 1} v_p\big(F'(t)\big) \qquad (F \in \mathbb{Z}_{(p)}[t]), \tag{3.1}$$

because $\Delta^b G(0)$ is an integer combination of values $F'(t)$ at odd t. For odd t we have $t - x_m = (t+2J) - 2m$, so for a level p^e the indices m with $p^e \mid t - x_m$ form one residue class $m_0(t)$ modulo p^e among the K consecutive integers $0,\dots,K-1$. Put

$$\mathrm{lev}^*(p) := \#\{e \ge 1 : p^e \le h\}.$$

3.1. **The atom expansion and the class count.** Let $V_{m,i} := x_m^i$ ($m < K$, $i < h$), $\mathcal{P}_{ij} := \mathcal{M}(R_{t^{i+j}})$ and $\gamma_m(X) := -z^{x_m}(X - P(x_m))/D'(x_m)$. By Lemma 2.1, $(E_{i+j}) = \mathcal{P} + V^{\mathsf{T}}\,\mathrm{diag}(\gamma)V$.

Lemma 3.1 (Lemma 7′, atom expansion)**.**

$$\Delta_h(X) = \sum_{|I|=|I'|=|S|} \pm \det \mathcal{P}_{I^c I'^c}\, \det V_{S,I}\, \det V_{S,I'} \prod_{m\in S} \gamma_m(X),$$

over $I, I' \subseteq \{0,\dots,h-1\}$ and $S \subseteq \{0,\dots,K-1\}$. Moreover $\det V_{S,I} \in \mathrm{Vand}(x_S)\mathbb{Z}$, and $\prod_{m\in S}|D'(x_m)| = \mathrm{Vand}(x_S)^2\,\mathrm{cross}(S)$ with $\mathrm{cross}(S) := \prod_{m\in S, l\notin S}|x_m - x_l|$.

Proof. Expand $\det(\mathcal{P}+M)$ by the complementary minors of $\mathcal{P}$ and $M = V^{\mathsf{T}}\operatorname{diag}(\gamma)V$, and each minor $\det M_{II'}$ by the Cauchy–Binet formula. A generalised Vandermonde determinant is $\operatorname{Vand}(x_S)$ times a Schur polynomial with integer coefficients. The last identity splits $\prod_{l\neq m}|x_m-x_l|$ into $l\in S$ and $l\notin S$. □

For an odd level d put $w_d(m) := 2[\,|x_m|>d\,]$, let $C_c := \{m<K : m\equiv c \bmod d\}$, $n_c := |C_c|$, and

$$A_K(d) := \sum_{c \bmod d} \max_{S_c\subseteq C_c}\Big[|S_c|(n_c-|S_c|)+\sum_{m\in S_c} w_d(m)\Big].$$

Lemma 3.2 (Lemma 8′, per-level bound)**.** *For every odd prime p, $n_p \le \sum_{e\ge1,\,p^e<K} A_K(p^e)$.*

Proof. $\mathcal{P}$ is $\mathfrak{p}$-integral by (3.1). In the term of Lemma 3.1 indexed by (I,I',S), the factors $\det V_{S,I}\det V_{S,I'}$ cancel $\operatorname{Vand}(x_S)^2$, so by Lemma 3.1 the term has valuation at least $-v_p(\operatorname{cross}(S))+\sum_{m\in S} v(X-P(x_m))$, using that z^{x_m} is a unit. Here $v_p(\operatorname{cross}(S)) = \sum_{e\ge1}\#\{(k,l) : k\in S,\ l\notin S,\ p^e\mid k-l\}$ since $x_k-x_l=2(k-l)$, and $v(X-P(x_k))\ge -2\max_{n\text{ odd},|n|<|x_k|} v_p(n) \ge -2\sum_{e\ge1}[\,p^e<|x_k|\,]$. At level $d=p^e$ the pairs counted are those inside residue classes, $\sum_c |S\cap C_c|(n_c-|S\cap C_c|)$, and $[\,p^e<|x_k|\,] = \frac12 w_d(k)$. Bounding the maximum over S level by level and class by class gives the claim. Levels $p^e\ge K$ contribute nothing: the classes are singletons and $|x_m|\le h<p^e$. □

Lemma 3.3 (Lemma 8A′, class count)**.** *For odd d with $K/3\le d<K$ we have $A_K(d)=3(K-d)$, and $A_K(d)=0$ for $d\ge K$.*

Proof. For $d\ge K$ all classes are singletons and all weights vanish. Let $K/3\le d<K$. Then $w_d(m)=0$ if and only if $|m-J|\le (d-1)/2$, a block of d consecutive indices inside $[0,K)$; so every class contains exactly one atom of weight 0, and the others have weight 2. A class with n atoms contributes $V(n) := \max_t[t(n-t)+2\min(t,n-1)]$ (take the weight-0 atom last), and $V(1)=0$, $V(2)=3$, $V(3)=6$. Since $3d\ge K$, classes have at most three atoms. If $K/3\le d\le K/2$ there are $K-2d$ classes of size 3 and $3d-K$ of size 2, with total $6(K-2d)+3(3d-K)=3(K-d)$. If $K/2<d<K$ there are $K-d$ pairs and $2d-K$ singletons, with total $3(K-d)$. □

3.2. **The plateau: a bordered Lagrange Gram identity.** In [14] the number of poles equals the size h of the determinant, and the Lagrange basis has the right dimension. Here $K=h+1$, and the K Lagrange polynomials $f_i := D/(t-x_i)$ have degree h: they span the polynomials of degree $\le h$, one dimension too many. The adjugate of the $(h+1)$-Gram matrix repairs this.

Lemma 3.4 (Lemma B, bordered identity)**.** *Let* $G^f(X) := (\Lambda_X(f_i f_j))_{i,j<K}$. *Then*

$$\Delta_h(X) = (-1)^{K(K-1)/2}\,\frac{\mathbf{1}^{\mathsf{T}}\operatorname{adj}\big(G^f(X)\big)\mathbf{1}}{\prod_{i<K} D'(x_i)}.$$

Proof. Let $H_{h+1} := (E_{a+b})_{a,b\le h}$. Then $\Delta_h = \operatorname{adj}(H_{h+1})_{hh}$. Let C be the coefficient matrix of the f_i (row i lists the coefficients of $t^0,\dots,t^h$ in f_i). Then $G^f = CH_{h+1}C^{\mathsf{T}}$, so $\operatorname{adj}(G^f) = (\det C)^2 C^{-\mathsf{T}}\operatorname{adj}(H_{h+1})C^{-1}$ and $\operatorname{adj}(H_{h+1}) = (\det C)^{-2}C^{\mathsf{T}}\operatorname{adj}(G^f)C$. Column h of C is $\mathbf{1}$, since every f_i is monic of degree h; hence $\Delta_h = (\det C)^{-2}\mathbf{1}^{\mathsf{T}}\operatorname{adj}(G^f)\mathbf{1}$. Finally $(f_i(x_k))_{i,k} = \operatorname{diag}(D'(x_i)) = CW$ with $W = (x_k^a)_{a,k}$, and $\prod_i D'(x_i) = (-1)^{K(K-1)/2}(\det W)^2$, so $(\det C)^2 = (-1)^{K(K-1)/2}\prod_i D'(x_i)$. □

Lemma 3.5 (Lemma 9′, plateau bound)**.** *For every odd prime* p, $n_p \le 2h\,\mathrm{lev}^*(p) + v_p(h!)$.

Proof. Put $N_e(i) := \#\{m < K : m \ne i,\ p^e \mid m-i\}$ and $N(i) := \sum_e N_e(i) = v_p(D'(x_i))$. Note that $N_e(i) \ge 1$ needs $p^e \le K-1 = h$. We first prove the *row claim*

$$v\big(G^f_{ij}\big) \ge N(i) - 2\,\mathrm{lev}^*(p) \qquad (i,j<K). \tag{3.2}$$

Fix an odd t and a level $d = p^e$, and let $m_0 = m_0(t)$. The class of i modulo d has $N_e(i)+1$ elements in $[0,K)$, and classes modulo d in $[0,K)$ differ in size by at most one, so the class m_0 has at least $N_e(i)$ elements.

Off-diagonal entries ($i \ne j$). Here $f_i f_j/D = F := \prod_{m\ne i,j}(t-x_m)$, so $G^f_{ij} = \mathcal{M}(F)$. Each term of $F'(t)$ omits one factor. At level d the factors divisible by d are those with $m \equiv m_0$, $m \ne i,j$; after the omission at least $N_e(i)-2$ remain, whether $m_0 \equiv i$ or not. Summing over levels, every term has $v_p \ge \sum_e (N_e(i) - 2[\,N_e(i)\ge 1\,]) \ge N(i) - 2\,\mathrm{lev}^*(p)$, and (3.1) gives (3.2).

Diagonal entries. Put $g := \prod_{m\ne i}(t-x_m)$. Then $f_i^2/D = g/(t-x_i) = Q + g(x_i)/(t-x_i)$ with $Q := (g(t)-g(x_i))/(t-x_i)$ and $g(x_i) = D'(x_i)$. By Lemma 2.1 the pole part contributes $-D'(x_i)z^{x_i}(X - P(x_i))$, of valuation at least $N(i) - 2\#\{e : p^e < |x_i|\} \ge N(i) - 2\,\mathrm{lev}^*(p)$, since $|x_i| \le h$. For Q, list the roots x_m, $m \ne i$, of g as $\alpha_1,\dots,\alpha_h$ in order of decreasing $v_p(m-i)$; telescoping gives

$$Q(t) = \sum_{r=1}^{h}\prod_{s<r}(x_i-\alpha_s)\prod_{s>r}(t-\alpha_s),$$

and for every level e the $\alpha_s = x_m$ with $p^e \mid m-i$ form an initial segment of length $N_e(i)$ (note $x_i - x_m = 2(i-m)$). Fix a term r of $Q'(t)$, which omits one more factor of the second product, and a level d. If $r > N_e(i)$, the first product has $N_e(i)$ factors divisible by d. If $r \le N_e(i)$ and $m_0 \equiv i$, it has $r-1$, and the remaining $N_e(i)-r$ class members sit at positions $s > r$, at most one of them omitted: in total at least $N_e(i)-2$. If $r \le N_e(i)$ and $m_0 \not\equiv i$, the whole class m_0 (at least $N_e(i)$ members, all with $v_p(m-i) < e$)

sits at positions $s > N_e(i) \geq r$: in total at least $N_e(i) - 1$. So (3.2) holds for Q too, by (3.1).

The minors. $\mathbf{1}^{\mathsf{T}} \operatorname{adj}(G^f)\mathbf{1} = \sum_{a,b} \pm M_{ba}$, where M_{ba} is the minor with row b and column a deleted. By (3.2) and the Leibniz expansion, $v(M_{ba}) \geq \sum_{i \neq b}(N(i) - 2\operatorname{lev}^*(p)) \geq \sum_i N(i) - 2h\operatorname{lev}^*(p) - \max_b N(b)$. Lemma 3.4 and $v_p(\prod_i D'(x_i)) = \sum_i N(i)$ give $v(\Delta_h) \geq -2h\operatorname{lev}^*(p) - \max_b N(b)$. Finally $N(b) = v_p(b!\,(h-b)!) \leq v_p(h!)$. □

Remark 3.6. On the exact profiles computed for $h = 16, \ldots, 256$ the bound of Lemma 3.5 is never violated, and it is attained at most single-level plateau primes; for example at $h = 128$ the primes $13, 17, 19, 23, 31, 37, 41, 43$ have $n_p = 256 + v_p(128!)$ exactly. The row claim (3.2) was checked entrywise at $h = 16, 24, 32, 40$ and $p \leq 13$, with minimum slack 0.

3.3. **The sum over the odd primes.**

Theorem 3.7 (Theorem A′)**.** *For every $\varepsilon > 0$ and all large even h, $\sum_{p \text{ odd}} n_p \log p \leq (\frac{4}{3} + \varepsilon)h^2$.*

Proof. For $p < K/3$ use Lemma 3.5:

$$\sum_{p<K/3} \big(2h\operatorname{lev}^*(p) + v_p(h!)\big)\log p \leq 2h\big(\theta(K/3) + \psi(h) - \theta(h)\big) + \log h!$$
$$= \tfrac{2}{3}h^2 + o(h^2)$$

by the prime number theorem. For $K/3 \leq p < K$ we have $p^2 \geq K^2/9 \geq K$ for $h \geq 8$, so only the level $e = 1$ occurs, and Lemmas 3.2 and 3.3 give $n_p \leq 3(K - p)$. By the prime number theorem and partial summation, $\sum_{K/3 \leq p < K} 3(K-p)\log p = 3h^2 \int_{1/3}^1 (1-x)\,dx + o(h^2) = \frac{2}{3}h^2 + o(h^2)$. For $p \geq K$, $n_p \leq 0$ by Lemma 3.2. □

The finite sums of these bounds, divided by h^2, are 1.3380, 1.3350 and 1.3337 at $h = 1442$, 20836 and 875032. The prime 5 needs no separate treatment: z and $1 - y = 4z$ are units at $\mathfrak{p} = (\sqrt{5})$, so every step above holds there with $v(5) = 1$.

4. Denominators II: the prime 2

Let v be the 2-adic valuation of $\mathbb{K}$; 2 is inert, $v(2) = 1$, $v(z) = 0$ and $v(1 - y) = 2$.

Rescaling. Put $\tau := t/2$. The atoms become $\tau = s + \frac{1}{2}$ ($s \geq 0$), the poles $\xi_k := k - J \in \mathbb{Z}$, and $D(t) = 2^K \tilde{D}(\tau)$ with $\tilde{D}(\tau) := \prod_{k<K}(\tau - \xi_k)$. With $\tilde{\mathcal{M}}(F) := \sum_{s \geq 0} z^{2s+1} F'(s + \frac{1}{2})$ and $\tilde{\Lambda}(g) := \tilde{\mathcal{M}}(g/\tilde{D})$ we get $\Lambda(t^m) = 2^{m-K-1}\tilde{\Lambda}(\tau^m)$, hence $\Delta_h = 2^{-3h}\tilde{\Delta}_h$ with $\tilde{\Delta}_h := \det(\tilde{\Lambda}_X(\tau^{i+j}))$, since $\sum_{i<h} 2i - h(K+1) = -3h$.

Anchored basis. Put $a_i := \prod_{l<i}(\tau - \xi_l)$ and $b_j := \prod_{l<j}(\tau - \xi_{K-1-l})$. Both families are monic of degrees i and j, so $\det(G_{ij}) = \tilde{\Delta}_h$ for $G_{ij} := \tilde{\Lambda}_X(a_i b_j)$.

Lemma 4.1 (Lemma 10′)**.** *Let $i, j < h$.*

(L) *If $\pi := i + j - K \geq 1$, then $v(G_{ij}) \geq -\pi - \lfloor \log_2 \pi \rfloor - 2$. If $\pi = 0$, then $G_{ij} = 0$.*

(U) *If $N_0 := K - 1 - i - j \geq 1$, then $v(G_{ij}) \geq N_0 - \lfloor \log_2 N_0 \rfloor$. If $N_0 = 0$, then $v(G_{ij}) = 2$.*

Consequently $n_2 \leq h\lfloor \log_2 h \rfloor + 3h$.

Proof. (L) The products a_i and b_j cover the index ranges $[0, i-1]$ and $[K-j, K-1]$, which overlap in π indices, so $a_i b_j/\tilde{D} = F := \prod_{l=K-j}^{i-1}(\tau - \xi_l) = 2^{-\pi}H(2\tau)$ with $H(U) := \prod_l (U - 2\xi_l) \in \mathbb{Z}[U]$. So $F'(s+\frac{1}{2}) = 2^{1-\pi}H'(2s+1)$, and by (2.1) (with s in place of $(t-1)/2$), $\tilde{\mathcal{M}}(F) = z\sum_{b<\pi} \Delta^b G(0) y^b/(1-y)^{b+1}$, $G(s) = 2^{1-\pi}H'(2s+1)$. The Newton coefficients of $s \mapsto (2s+1)^k$ are $\Delta^b[(2s+1)^k](0) = \sum_{j \geq b} \binom{k}{j} 2^j\, b!\, S(j,b)$, of valuation $\geq b + v_2(b!)$, while $y^b/(1-y)^{b+1}$ has valuation $-2(b+1)$. Hence $v(G_{ij}) \geq 1 - \pi + \min_{b<\pi}(-2 - s_2(b)) \geq -\pi - \lfloor \log_2 \pi \rfloor - 2$. There is no X-part. If $\pi = 0$ then $F = 1$ and $\tilde{\mathcal{M}}(1) = 0$.

(U) Now the ranges leave a gap of $N_0 + 1$ indices, and with $\zeta_l := \xi_i + l$,

$$\frac{a_i b_j}{\tilde{D}} = \prod_{l=0}^{N_0} \frac{1}{\tau - \zeta_l} = \frac{1}{N_0!} \sum_{l=0}^{N_0} (-1)^{N_0 - l} \binom{N_0}{l} \frac{1}{\tau - \zeta_l}.$$

For an integer ζ, $\tilde{\mathcal{M}}(1/(\tau - \zeta)) = -4\sum_s z^{2s+1}/(2s+1-2\zeta)^2 = -4z^{2\zeta}(X - S(\zeta))$ (at $X = T$), where $S(\zeta+1) - S(\zeta) = -z^{-1-2\zeta}/(2\zeta+1)^2$; so $S(\zeta)$ has only odd denominators and $v(S(\zeta)) \geq 0$. With $z^{2\zeta_l} = z^{2\xi_i} y^l$ and $S(\zeta_l) = S(\xi_i) - \sum_{r<l} z^{-1-2\xi_i-2r} g(r)$, $g(r) := (s_0 + 2r)^{-2}$, $s_0 := 2\xi_i + 1$,

$$G_{ij} = -\frac{4}{N_0!}\Big[z^{2\xi_i}(y-1)^{N_0}\big(X - S(\xi_i)\big) + z^{-1}\sum_l (-1)^{N_0-l}\binom{N_0}{l}\sum_{r<l} y^{l-r} g(r)\Big].$$

The first part has valuation $2N_0$, since $y - 1 = -4z$. In the second, write $y^{l-r} = y\sum_e \binom{l-r-1}{e}\eta^e$ with $\eta := y - 1$ ($v(\eta) = 2$) and $g(r) = \sum_m \gamma_m \binom{r}{m}$; expanding $g(r) = s_0^{-2}\sum_k (-1)^k (k+1)(2/s_0)^k r^k$ (convergent in $\mathbb{Z}_2$) and using $\Delta^m r^k(0) = m!\, S(k,m)$ gives $v_2(\gamma_m) \geq m + v_2(m!) = 2m - s_2(m)$. The identities $\sum_{r<l}\binom{r}{m}\binom{l-1-r}{e} = \binom{l}{m+e+1}$ and $\sum_l (-1)^{N_0-l}\binom{N_0}{l}\binom{l}{M} = [M = N_0]$ collapse the second part to $z^{-1} y \sum_{m+e=N_0-1} \gamma_m \eta^e$, of valuation $\geq 2N_0 - 3 - \lfloor \log_2 N_0 \rfloor$. With $v_2(N_0!) = N_0 - s_2(N_0)$ this gives $v(G_{ij}) \geq N_0 - 1 + s_2(N_0) - \lfloor \log_2 N_0 \rfloor \geq N_0 - \lfloor \log_2 N_0 \rfloor$; the X-part has valuation $N_0 + 2 + s_2(N_0)$. If $N_0 = 0$ then $G_{ij} = -4z^{2\xi_i}(X - S(\xi_i))$, of valuation exactly 2.

Assignment. In all cases $v(G_{ij}) \geq K - i - j - \lfloor \log_2 h \rfloor - 2$, since $\pi, N_0 \leq h$. By the Leibniz expansion, $v(\tilde{\Delta}_h) \geq \sum_{i<h}(K - i - \sigma(i)) - h(\lfloor \log_2 h \rfloor + 2) = -h\lfloor \log_2 h \rfloor$ for every permutation σ, and $n_2 = -v(\Delta_h) = 3h - v(\tilde{\Delta}_h)$. □

The mechanism is $v_2(1 - y) = 2$: both triangles (L) and (U) then have slope 1 in $i + j$, and the permutation sum of the bound is constant. In [14] the corresponding quantity is $v_2(1 - q^{-2}) = 3$, which costs $\frac{1}{4}\log 2$ per

h^2. Section 8 shows what happens when it is 1. The exact values $n_2 = 54, 96, 142, 240, 350, 576, 830$ at $h = 16, 24, 32, 48, 64, 96, 128$ lie below the bound of Lemma 4.1 $(112, 168, 256, 384, 576, 864, 1280)$, and $n_2/(h \log_2 h)$ rises from 0.84 to 0.94 for $h \le 256$.

5. The size of $\Delta_h(T)$: Theorem B′

5.1. **The equilibrium problem and weak duality.** Put

$$\Psi(x) := \tfrac{1}{2} \int_{x-1}^{x+1} \log|s|\, ds, \qquad v(x) := x \log z - \Psi(x).$$

Ψ is even, increasing on $[0, \infty)$, and $\Psi(0) = -1$. The function v is continuous, and $v'(x) = \log z - \frac{1}{2} \log|(x+1)/(x-1)|$ for $x \neq \pm 1$. For $x > 0$, $x \neq 1$, the logarithm is positive, so $v' < \log z < 0$; for $-1 < x \le 0$, $v'(x) = \log z + \operatorname{artanh}|x| < 0$ if and only if $|x| < \tanh(\log(1/z)) = 0.8945\ldots$. Hence

$$v \text{ is strictly decreasing on } (-0.894, \infty) \supset [-1/h, \infty) \qquad (h \ge 2). \tag{5.1}$$

For a finite positive measure ρ with compact support put $U^\rho(x) := \int \log|x - s|\, d\rho(s)$ and $E(\rho) := \iint \log|x - s|\, d\rho(x)\, d\rho(s)$.

Call μ *admissible* if it is a density with $0 \le \mu \le 1$, mass 1 and $\operatorname{supp} \mu \subset [0, 2]$. For admissible μ and $\lambda \in \mathbb{R}$ put

$$F_\mu := 2U^\mu + v, \qquad \Phi(\mu, \lambda) := -E(\mu) + \lambda + \int_0^\infty (F_\mu - \lambda)_+\, dx.$$

If ρ is a density with $0 \le \rho \le 1$, mass 1 and compact support, then $E(\rho - \mu) \le 0$ [24, Lemma I.1.8], i.e. $E(\rho) \le 2 \int U^\mu\, d\rho - E(\mu)$, so

$$E(\rho) + \int v\, d\rho \le -E(\mu) + \lambda + \int (F_\mu - \lambda)\, d\rho. \tag{5.2}$$

The supremum of the left side over such ρ supported in $[0, \infty)$ is the constrained equilibrium constant C; by (5.2), $C \le \Phi(\mu, \lambda)$.

5.2. **The theorem.**

Theorem 5.1 (Theorem B′)**.** *Let μ be admissible and $\lambda \ge -2.96317$, and suppose that $F_\mu - \lambda \le -1.13 - 0.94(x - 4)$ on $[4, \infty)$. Then for every even $h \ge 100$,*

$$\log|\Delta_h(T)| \le h^2\, \Phi(\mu, \lambda) + E_{\mathrm{an}}(h),$$

where $c_1(h) := \log(2h) + 1 + \log(2 + \log(2h + 1))$ and

$$E_{\mathrm{an}}(h) := 2h \log(4h) + h \log 4 + (2h+1) \log 3 + h\, c_1(h) + 7.742h - h \log h + 1.$$

In particular $\limsup_{h \to \infty} h^{-2} \log|\Delta_h(T)| \le \Phi(\mu, \lambda)$.

The proof follows the proof of Theorem B in [14] in the first two steps and then departs from it: the poles now lie on both sides of the atoms, which is handled by an exact Jensen-type bound (step 4), and the discretisation spreads the particles into cells (step 5) instead of using a lattice energy inequality.

Proof. Write $\phi := 1/D$.

1. Node expansion. For $a, b \in \mathbb{Q}[t]$, $(ab\phi)' = a'b\phi + ab'\phi + ab\phi'$, so $\Lambda(ab) = \sum_{t\ \mathrm{odd}\geq 1}(a,a')(t)\, M_t\, (b,b')(t)^{\mathsf{T}}$ with $M_t := z^t\left(\begin{smallmatrix}\phi' & \phi\\ \phi & 0\end{smallmatrix}\right)(t)$. Let $H^{(t_{\max})}$ be the Hankel matrix of the sums truncated at $t \leq t_{\max}$. Each node t carries a *value slot* (the row $(t^i)_{i<h}$) and a *derivative slot* (the row $(it^{i-1})_{i<h}$). By the Cauchy–Binet formula, $\det H^{(t_{\max})}$ is a sum over *patterns* $n = (n_t)$, $n_t \in \{0,1,2\}$, $\sum_t n_t = h$, and over pairs (S, S') of slot sets with n_t slots at node t, of $\pm \det E_S \det \mathbb{M}_{S,S'} \det E_{S'}$, where E_S is the matrix of the rows in S and $\mathbb{M}$ is block diagonal with blocks M_t. There are at most 4^h pairs (S, S') per pattern.

2. Slot weights. $|\det \mathbb{M}_{S,S'}| \leq \prod_t (z^t m_t)^{n_t}$ with $m_t := \max(|\phi(t)|, |\phi'(t)|)$ (for $n_t = 2$ note $|\det M_t| = z^{2t}\phi(t)^2$). The K numbers $t - x_m$ are distinct odd integers, so $|\phi'/\phi|(t) \leq \sum_m 1/|t - x_m| \leq 2\sum_{k=0}^{h} 1/(2k+1) \leq 2 + \log(2h+1)$, and $m_t \leq |\phi(t)|(2 + \log(2h+1))$.

3. Vandermonde determinants with derivative rows. $|\det E_S| \leq (4h)^h \prod_{t<t'} |t - t'|^{n_t n_{t'}}$. Indeed, replace the derivative-only slots $s_1, \ldots, s_r$ of S by value rows at free variables; the result is a confluent Vandermonde determinant $\pm\prod_{a<b}(a-b)^{n_a n_b}$, and $\det E_S$ is obtained by applying $\partial_{s_1}\cdots\partial_{s_r}$ and evaluating at the nodes. By induction, after k derivatives the expression is a sum of at most $(h-1)^k$ terms, each a product of k exponents (each ≤ 2) times $\prod (a-b)^{e'_{ab}}$ with $0 \leq e'_{ab} \leq n_a n_b$. At the nodes $|a-b| \geq 2 > 1$, so each term is at most $2^k \prod |a-b|^{n_a n_b}$, and $(2(h-1))^r \leq (4h)^h$.

4. The pole product. For odd t the numbers $t - x_m$ are the odd integers in $[t-h, t+h]$. For every odd u, $\log|u| \geq \frac{1}{2}\int_{u-1}^{u+1} \log|s|\, ds$, by concavity of log on $(0,\infty)$ and on $(-\infty, 0)$ (the singular endpoint is integrable). The intervals $[u-1, u+1]$ tile $[t-h-1, t+h+1]$, so with $x := t/h$ and $s = u/h$,

$$\begin{aligned}\log|D(t)| &\geq \tfrac{1}{2}\int_{t-h-1}^{t+h+1} \log|u|\, du\\ &= (h+1)\log h + h\Psi(x) + \tfrac{h}{2}\Big[\int_{x+1}^{x+1+1/h} + \int_{x-1-1/h}^{x-1}\Big]\log|s|\, ds.\end{aligned}$$

Each bracketed integral is over an interval of length $1/h$, hence at least $\int_{-1/(2h)}^{1/(2h)} \log|s|\, ds = -(\log(2h)+1)/h$. So $\log|D(t)| \geq (h+1)\log h + h\Psi(x) - \log(2h) - 1$ and, with step 2,

$$t\log z + \log m_t \leq h\, v(t/h) - (h+1)\log h + c_1(h). \tag{5.3}$$

By steps 1–4, the pattern n contributes at most $4^h(4h)^{2h}e^{D(n)}$ with $D(n) := \sum_{t\neq t'} n_t n_{t'} \log|t-t'| + \sum_t n_t\big(h\, v(t/h) - (h+1)\log h + c_1(h)\big)$.

5. Discrete to continuum. Spread the n_t slots of node t uniformly over $I_t := [(t-2)/h, t/h]$: the density $\rho_n := \sum_t \frac{n_t}{2}\mathbf{1}_{I_t}$ satisfies $0 \leq \rho_n \leq 1$, has mass 1, and is supported in $[-1/h, \infty)$. By (5.1), $v(t/h) \leq v$ on I_t, so $\sum_t n_t h\, v(t/h) \leq h^2 \int v\, d\rho_n$. For $t \neq t'$ with $|t - t'| = 2d$, writing $x =$

$(t-2+2a)/h$, $a\in[0,1]$,

$$h^2\iint_{I_t\times I_{t'}}\log|x-s|\,d\rho_n\,d\rho_n = n_t n_{t'}\big(\log|t-t'|-\log h-\operatorname{gap}(d)\big),$$

$$\operatorname{gap}(d):=\log d-\int_0^1\int_0^1\log|d+a-a'|\,da\,da',$$

and $0\le\operatorname{gap}(d)\le 0.1137/d^2$ (with equality at $d=1$ up to rounding). A cell with itself gives $n_t^2(\log 2-\log h-\frac{3}{2})$. Summing, and using $n_{t'}\le 2$ and $\sum_t n_t^2\le 2h$,

$$\sum_{t\ne t'} n_t n_{t'}\log|t-t'|\le h^2E(\rho_n)+h^2\log h+0.748h+1.614h$$

($4\cdot 0.1137\,\zeta(2)\le 0.748$ and $2(\frac{3}{2}-\log 2)\le 1.614$). The terms in $\log h$ combine to $h^2\log h-h(h+1)\log h=-h\log h$, so

$$D(n)\le h^2J(\rho_n)-h\log h+h\,c_1(h)+2.362h,\qquad J(\rho):=E(\rho)+\int v\,d\rho.$$

6. Dual, the cell below 0*, and far nodes.* By (5.2), $h^2J(\rho_n)\le h^2(-E(\mu)+\lambda)+h^2\int(F_\mu-\lambda)\,d\rho_n$. On $[0,4]$ the integral is at most $\int_0^4(F_\mu-\lambda)_+dx$, since $\rho_n\le 1$. On $[-1/h,0]$, which only the cell I_1 meets, we have $U^\mu(x)\le\log(2+1/h)$ (as $\operatorname{supp}\mu\subset[0,2]$) and $v(x)\le|\log z|/h-\Psi(0)=1+1.4437/h$; hence $F_\mu\le 2.4108$ and $F_\mu-\lambda\le 2.4108+2.96317\le 5.374<5.38$ for $h\ge 100$, and the contribution to h^2J is at most $5.38h$. On $[4,\infty)$, by hypothesis, a slot at a node $t\ge 4h+3$ contributes at most $-a_t$ with $a_t:=h(1.13+0.94((t-2)/h-4))\ge 1.13h$; the node $4h+1$, whose cell straddles 4, has $F_\mu-\lambda<0$ on its part beyond 4. So for every pattern

$$h^2J(\rho_n)\le h^2\Phi(\mu,\lambda)+5.38h-\sum_{t\ge 4h+3}n_t a_t.$$

The $2h+1$ odd nodes $t\le 4h+1$ carry at most 3^{2h+1} patterns, and the nodes beyond contribute at most $\prod_{t\ge 4h+3}(1+e^{-a_t}+e^{-2a_t})\le\exp\big(2e^{-1.13h}/(1-e^{-1.88})\big)\le e$. Collecting,

$$\begin{aligned}\log|\det H^{(t_{\max})}|\le{}& h\log 4+2h\log(4h)+(2h+1)\log 3\\&+h^2\Phi(\mu,\lambda)+5.38h-h\log h+h\,c_1(h)+2.362h+1,\end{aligned}$$

uniformly in $t_{\max}$. As $t_{\max}\to\infty$ the truncated matrices converge entrywise to $(\Lambda(t^{i+j}))$, and the determinant is continuous. □

Remark 5.2. For every even $h\ge 2$ the same bound holds with 7.742 replaced by 9.592: for any h, the bound $U^\mu(x)\le\int_1^2\log(s+1)\,ds=0.9095\ldots$ for $x\in[-1/h,0]$ (because $\mu\le 1$ and $\operatorname{supp}\mu\subset[0,2]$) gives $F_\mu-\lambda\le 7.23$ on $[-1/h,0]$. Only the $O(h)$ term changes.

5.3. **The certificate.**

Theorem 5.3. *There is an admissible pair (μ, λ) satisfying the hypotheses of Theorem 5.1 with*

$$\Phi(\mu, \lambda) \le -1.3592161834586374.$$

The proof is a finite computation in ball arithmetic (Arb [11], through python-flint [27]) at 128 bits. The density μ is constant on N equal cells of $[0, 2]$, with rational heights in $[0, 1]$ and mass exactly 1 (checked in exact rational arithmetic). The energy $E(\mu)$ is computed from the exact double antiderivative $\frac{1}{2}u^2 \log|u| - \frac{3}{4}u^2$ as a Toeplitz sum over cell offsets. The grid has step d/r ($d = 2/N$ the cell width, r even, so cell midpoints are grid points), and on each grid interval $I = [a, b]$:

- the three cells nearest to I are treated exactly: each cell potential is unimodal with its minimum at the cell midpoint, so its maximum on I is at an endpoint that can be chosen exactly;
- the far field $f = 2\sum_{\text{far}} \mu_j U_j$ (every far cell is at distance $\ge d$ from I) satisfies $f'' \ge -4/d$, because $f'' = -2\int_{\text{far}} \mu(s)(x-s)^{-2} ds$ and $\mu \le 1$; and $v''(x) = 1/(x^2-1)$ is $\ge -1/(1-b^2)$ for $b < 1$ and ≥ 0 for $a > 1$. A function g with $g'' \ge -\kappa$ on I satisfies $\sup_I g \le \max(g(a), g(b)) + \kappa(b-a)^2/8$; this is applied to $f + v$ with $\kappa = 4/d + 1/(1-b^2)$ (if $b < 1$) or $\kappa = 4/d$ (if $a > 1$). On the two intervals touching $x = 1$, v is bounded by $v(a)$;
- $\int (F_\mu - \lambda)_+ \le \sum_I |I| \max(S_I - \lambda, 0)$, with S_I the resulting upper bound on I; on $[4, \infty)$, $F_\mu(x) \le 2\log x + x\log z - \log(x-1)$, which is $< \lambda$ and decreasing there.

A first version with per-cell maxima everywhere loses the cancellation between neighbouring cells and gives only -1.313 at $N = 100$. Table 2 lists the results; the last row proves Theorem 5.3.

N	r	$-E(\mu)$	λ	$\int (F_\mu - \lambda)_+ \le$	certified Φ
200	16	1.21902586	-2.96219282	0.38673	-1.35643705
400	32	1.21902867	-2.96297388	0.38528	-1.35866119
800	32	1.21902407	-2.96310589	0.38505	-1.35903096
1600	32	1.21902464	-2.96316943	0.384929	-1.35921618345864

TABLE 2. Dual certificates for Theorem 5.1 (decimal values rounded; λ is a rational number). For $N = 1600$ the certified Arb ball is $[-1.35921618345864 \pm 2.6 \cdot 10^{-15}]$, with upper endpoint -1.3592161834586374.

The side conditions of Theorem 5.1 hold for the certified pair: $\lambda = -2.96316943\ldots \ge -2.96317$; at $x = 4$, $2\log x + x\log z - \log(x-1) - \lambda = -1.137\ldots$, and the slope of this bound on $[4, \infty)$ is at most $2/x + \log z \le -0.94$.

Remark 5.4 (the equilibrium). An active-set quadratic program (and two independent codes) give the primal value $C \approx -1.3594016$, with maximiser saturated ($\rho = 1$) on $[0, 0.5475]$, a free band up to 1.9525, and $\lambda^* \approx -2.96317$; so the certificate is within about $1.9 \cdot 10^{-4}$ of the optimum. The finite-h values $h^{-2} \log |\Delta_h(T)|$, computed in ball arithmetic, are -1.4379, -1.4195, -1.4093, -1.4015, -1.3930, -1.3876, -1.3815, -1.3781 at $h = 16$, 32, 48, 64, 96, 128, 192, 256. They increase towards the constant from below; fits of the form $a + b \log h/h \, (+c/h)$ give limits between -1.373 and -1.364, below C, so these data do not indicate that Theorem 5.1 is sharp. This costs nothing, since only the upper bound is used.

6. Nonvanishing: Lemma P′

6.1. **Setting.** Let $\ell \geq 7$ be prime, $h := 3\ell - 1$ (even) and $K = 3\ell$. The classes modulo ℓ are $C_\gamma = \{\gamma, \gamma + \ell, \gamma + 2\ell\}$, $\gamma < \ell$, all of size 3. Since $|m - i| < 3\ell < \ell^2$ there is a single level, and $N(i) = v_\ell(D'(x_i)) = 2$ for every i. Let $\mathfrak{l}$ be a prime of $\mathbb{K}$ above ℓ, $v := v_\mathfrak{l}$, and $\zeta := z^\ell \bmod \mathfrak{l}$; then $\zeta = z$ if ℓ splits ($\ell \equiv \pm 1 \bmod 5$) and $\zeta = \sigma(z) = -1/z$ if ℓ is inert. Since $\ell \nmid 10$, the elements z, y, $1 - y = 4z$ and $1 - \zeta^{\pm 2}$ are $\mathfrak{l}$-units.

Positions. $x_{\gamma + r\ell} = x_\gamma + 2r\ell$ with $x_\gamma = 2\gamma - 3\ell + 1 \in [-3\ell + 1, -\ell - 1]$: in each class the pole with $r = 0$ is large and negative ($|x| > \ell$), the pole with $r = 1$ is small ($|x| < \ell$), and the pole with $r = 2$ is large and positive. For odd t we have $\ell \mid t - x_m$ exactly for m in one class $C_{\gamma(t)}$, and the odd $t \geq 1$ with $\gamma(t) = \gamma$ form the progression $t = x_\gamma + \ell Y$, $Y \in \{3, 5, 7, \dots\}$. Put $t_\gamma := x_\gamma + 3\ell$.

Lemma 6.1 (Lemma P′)**.** *Let $c_k := [X^k]\Delta_h$. Then $v_\ell(c_0) = -2K$ and $v_\ell(c_k) \geq -2K + 2k$ for all k. Hence $\Delta_h(a/b) \neq 0$ for every $a/b \in \mathbb{Q}$ with $v_\ell(b) \leq 1$.*

Proof. 1. Lemma B. $\ell^{2K}\Delta_h(X) = (-1)^{K(K-1)/2} u_\ell \, \mathbf{1}^{\mathsf{T}} \operatorname{adj}(G^f(X))\mathbf{1}$ with $u_\ell := \ell^{2K} / \prod_i D'(x_i)$ an ℓ-unit.

2. The X-part. $B(f_i f_j) = 0$ for $i \neq j$, because $f_i f_j / D$ is a polynomial, and $B(f_i^2) = -D'(x_i) z^{x_i}$, of valuation exactly 2. So $G^f(X) = G^f(0) + X \operatorname{diag}(d_i)$ with $v(d_i) = 2$.

3. Integrality and block diagonality. By the row claim (3.2) with $N(i) - 2\,\mathrm{lev}^*(\ell) = 0$, $G^f(0)$ is $\mathfrak{l}$-integral. On each progression, $\mathcal{M}$ restricted to γ is $z^{t_\gamma} \sum_k z^{2\ell k} F'(t_\gamma + 2\ell k) = z^{t_\gamma} \sum_b \Delta^b G_\gamma(0) z^{2\ell b} / (1 - z^{2\ell})^{b+1}$, $G_\gamma(k) := F'(t_\gamma + 2\ell k)$, with $1 - z^{2\ell} \equiv 1 - \zeta^2$ a unit; so (3.1) holds progression by progression. Off the diagonal, $F = \prod_{m \neq i,j} (t - x_m)$ has on the progression γ exactly $3 - [\,\gamma = c(i)\,] - [\,\gamma = c(j)\,]$ factors divisible by ℓ ($c(i)$ is the class of i), and F' keeps all but one; so a unit is possible only if $c(i) = c(j) = \gamma$. On the diagonal, the polynomial part on a progression $\gamma \neq c(i)$ has valuation at least $3 - 1 = 2$: the class γ has three members, all after the initial segment of the telescoping order, and the derivative loses at most one. Hence $G^f(0) \bmod \mathfrak{l}$ is

block diagonal over the classes, each block receiving only its own progression and its own pole terms.

4. The blocks. Fix γ with members $m_r = \gamma + r\ell$, put $W(t) := \prod_{m\notin C_\gamma}(t - x_m)$ and $U_\gamma := \prod_{m\notin C_\gamma}(x_\gamma - x_m)$, so $W(t) \equiv U_\gamma \bmod \ell$ on the progression. Off the diagonal ($r \neq s$, q the third index) $F = W(t)(t - x_{m_q})$ and $F' \equiv W \equiv U_\gamma$. For the diagonal polynomial part, $Q = W\cdot(t + x_{m_r} - x_{m_a} - x_{m_b}) + p(x_{m_r})\,W[x_{m_r}, t]$ with $\{a, b\}$ the other indices, $p(t) = (t - x_{m_a})(t - x_{m_b})$ and $p(x_{m_r}) \equiv 0 \bmod \ell^2$; so $Q' \equiv U_\gamma$ too. In the progression sum, $\Delta^0 G_\gamma(0) \equiv U_\gamma$ and $\Delta^b G_\gamma(0) \equiv 0$ for $b \geq 1$, so it is $\equiv U_\gamma z^{t_\gamma}/(1 - z^{2\ell}) \equiv U_\gamma z^{x_\gamma}\zeta^3/(1 - \zeta^2)$. The pole terms at $X = 0$ are $D'(x)z^x P(x)$ with $\ell^2 P(x_{m_r}) \equiv \zeta, 0, -\zeta^{-1}$ for $r = 0, 1, 2$ (the only odd multiples of ℓ below $|x| \leq 3\ell - 1$ in absolute value are $\pm\ell$), $D'(x_{m_r})/\ell^2 \equiv U_\gamma \cdot (8, -4, 8)_r$, and $z^{x_{m_r}} \equiv z^{x_\gamma}\zeta^{2r}$. Hence

$$G^f(0)|_{C_\gamma} \equiv e_\gamma M, \qquad e_\gamma := U_\gamma z^{x_\gamma},$$

$$M := c\,\mathbf{1}\mathbf{1}^{\mathsf{T}} + \operatorname{diag}(8\zeta, 0, -8\zeta^3), \qquad c := \frac{\zeta^3}{1 - \zeta^2}.$$

5. The bordered determinant. For $M = \operatorname{diag}(d_0, 0, d_2) + c\mathbf{1}\mathbf{1}^{\mathsf{T}}$ we have $\det M = c\,d_0 d_2 = -64c\zeta^4$ and $\mathbf{1}^{\mathsf{T}}\operatorname{adj}(M)\mathbf{1} = d_0 d_2 = -64\zeta^4$, both units. For the block-diagonal reduction,

$$\mathbf{1}^{\mathsf{T}}\operatorname{adj}\Big(\bigoplus_\gamma e_\gamma M\Big)\mathbf{1} = (\det M)^{\ell-1}\,\mathbf{1}^{\mathsf{T}}\operatorname{adj}(M)\mathbf{1}\,\Big(\prod_\gamma e_\gamma^3\Big)\sum_\gamma e_\gamma^{-1}.$$

Modulo ℓ, $U_\gamma \equiv \prod_{m\not\equiv\gamma} 2(\gamma - m) \equiv 2^{3(\ell-1)}((\ell-1)!)^3 \equiv -1$ for every γ (Fermat, Wilson). So $\sum_\gamma e_\gamma^{-1} \equiv -z^{3\ell-1}\sum_{\gamma<\ell} z^{-2\gamma} = -z^{3\ell-1}(1 - z^{-2\ell})/(1 - z^{-2})$, and $1 - z^{-2\ell} \equiv 1 - \zeta^{-2}$ and $1 - z^{-2} = -4/z$ are units. Hence $\mathbf{1}^{\mathsf{T}}\operatorname{adj}(G^f(0))\mathbf{1}$ is an $\mathfrak{l}$-unit, and $v_\ell(c_0) = -2K$ by step 1.

6. The gap. $[X^k]\mathbf{1}^{\mathsf{T}}\operatorname{adj}(G^f(X))\mathbf{1}$ is a sum of products containing k factors d_i, so it has valuation $\geq 2k$. If $v_\ell(b) \leq 1$, the k-th term of $\Delta_h(a/b)$ has valuation $\geq -2K + 2k - k > -2K$ for $k \geq 1$. □

Every prime $\ell \geq 7$ works, split or inert. In [14] the partial sums reach past 3ℓ and the block constants are fixed rationals that must avoid ℓ, so finitely many primes are excluded; here only $n = \pm\ell$ enters. Steps 4–5 were checked entrywise modulo ℓ for $\ell = 7, 11, \ldots, 41$ (split and inert), and the conclusion on the exact Δ_h for $\ell = 7, \ldots, 29$.

7. Proof of Theorem 1.1

Suppose that $T = a/b$ with $a \in \mathbb{Z}$, $b \geq 1$. Let $\mathcal{H} := \{3\ell - 1 : \ell \geq 37 \text{ prime},\ \ell^2 \nmid b\}$, an infinite set of even integers $h \geq 110$. For $h \in \mathcal{H}$, Lemma 6.1 gives $\Delta_h \not\equiv 0$ and $F_h(a/b) \neq 0$, where $F_h \in \mathbb{Z}[X]$ is the primitive polynomial of Section 2, of degree $\leq h$. We have $\log|F_h(T)| = \log|\Delta_h(T)| + \sum_p n_p \log p$, and along $\mathcal{H}$:

- odd primes: $\limsup h^{-2}\sum_{p\text{ odd}} n_p \log p \leq \frac{4}{3}$ (Theorem 3.7);
- $p = 2$: $n_2 = O(h\log h)$ (Lemma 4.1);

- size: $\limsup h^{-2}\log|\Delta_h(T)| \le -1.3592161834586374$ (Theorems 5.1 and 5.3).

So $\limsup_{h\in\mathcal{H}} h^{-2}\log|F_h(T)| \le -0.0258828\ldots < 0$, contradicting Proposition 2.4. Hence $T = \chi_2(\sqrt{5}-2)$ is irrational, and so is $24T = \pi^2 - 18\log^2\varphi$ by (1.1). □

Remark 7.1 (effectivity). Every bound above is explicit at finite h except the prime number theorem in Theorem 3.7. At a single h one can instead evaluate the proved finite-h bound

$$\log|F_h(T)| \le h^2\Phi + E_{\mathrm{an}}(h) + B_{\mathrm{odd}}(h) + h(\lfloor\log_2 h\rfloor + 3)\log 2,$$

where $B_{\mathrm{odd}}(h) = \sum_{3\le p<K/3}(2h\,\mathrm{lev}^*(p) + v_p(h!))\log p + \sum_{K/3\le p<K}\min(2h + v_p(h!), 3(K-p))\log p$ is an exact finite sum. In exact arithmetic this bound is negative at every even h with $2432 \le h \le 40000$, and at every sampled h up to $8.75\cdot 10^5$, with $|\mathrm{bound}|/h^2$ increasing towards 0.0259. A proof that it stays negative for all larger h would need explicit error terms in the prime number theorem, which we have not carried out. As an effective consequence, if $T = a/b$ with $\log b \le 800$, a single $h = 3\ell - 1 \le 36000$ with $\ell \nmid b$ gives a contradiction. The exact scores $h^{-2}\log|F_h(T)|$ are much better than this bound, for example -0.066 at $h = 192$ (Table 6).

8. Two points

8.1. **The statement.** Let $n \ge 6$ be even and $P_n(X) := X^4 + nX^3 + 2X^2 - nX + 1$. It is irreducible with four real roots: two in $(0,1)$, which we call $z > w$, and $-1/z$, $-1/w$; moreover $w = (1-z)/(1+z)$, and $z \approx 1 - 2/n$, $w \approx 1/n$. Let $\mathcal{K} := \mathbb{Q}(z)$, a totally real biquadratic field with $\mathrm{Gal}(\mathcal{K}/\mathbb{Q}) = \{1, s, l, sl\}$, $s(z) = -1/z$, $l(z) = w$. Put $u := z - 1/z$; then $u^2 + nu + 4 = 0$, and $\mathcal{F} := \mathcal{K}^s = \mathbb{Q}(u) = \mathbb{Q}(\sqrt{n^2-16})$, on which l acts as the nontrivial automorphism, with $l(u) = 4/u$. With $\varepsilon_k := (k + \sqrt{k^2-4})/2$ we have $zw = \varepsilon_{n+2}^{-1}$ and $z/w = \varepsilon_{n-2}$. Put $T_z := \chi_2(z)$, $T_w := \chi_2(w)$; Landen's identity gives

$$T_n := T_z + T_w = \frac{\pi^2}{8} - \frac{1}{2}\log z\log w = \frac{\pi^2}{8} - \frac{1}{8}\log^2\varepsilon_{n+2} + \frac{1}{8}\log^2\varepsilon_{n-2}.$$

Definition 8.1. M1(n) is the statement: $T_n \notin \mathbb{Q}$, or $T_z - T_w \notin \sqrt{n^2-16}\,\mathbb{Q}$.

Equivalently, there is no $\alpha \in \mathcal{F}$ with $T_z = \alpha$ and $T_w = l(\alpha)$: such an α would have $\alpha + l(\alpha) \in \mathbb{Q}$ and $\alpha - l(\alpha) \in \sqrt{n^2-16}\,\mathbb{Q}$, and conversely. (Exchanging z and w only changes the sign of $T_z - T_w$.)

Theorem 8.2 (Theorem M1)**.** M1(n) *holds for every* $n \in \mathcal{N} := \{730, 740, 750, 760, 800, 1000\}$.

The proof is computer-assisted: for each $n \in \mathcal{N}$ separately it uses two dual certificates checked in interval arithmetic. It is a disjunction at six specific values of n, not a statement about a range of n, and it is not formalised.

The case $n = 730$ (Theorem 1.2). Here $z = 0.99726400650629232574\ldots$, $w = 0.00136987072554999983 52\ldots$, $\varepsilon_{732} = 366 + \sqrt{133955} = 1/(zw)$, $\varepsilon_{728} = 364 + 11\sqrt{1095} = z/w$ and $\sqrt{730^2 - 16} = 22\sqrt{1101}$. Numerically $T_{730} = 1.2246689334990568135\ldots$, $T_z = 1.2232990624878816026\ldots$ and $T_w = 0.0013698710111752109\ldots$. The fields are $\mathcal{K} = \mathbb{Q}(\sqrt{133955}, \sqrt{1095})$ and $\mathcal{F} = \mathbb{Q}(\sqrt{1101})$, and the primes ramified in $\mathcal{K}$ are $2, 3, 5, 73, 367$. The other instances are in Table 3; in each case $\mathcal{K} = \mathbb{Q}(\sqrt{d_+}, \sqrt{d_-})$ with $d_\pm$ the squarefree parts of $(n \pm 2)^2 - 4$.

n	$\varepsilon_{n+2} = 1/(zw)$	$\varepsilon_{n-2} = z/w$	$\sqrt{n^2-16}$	ramified in $\mathcal{K}$
730	$366 + \sqrt{133955}$	$364 + 11\sqrt{1095}$	$22\sqrt{1101}$	$2, 3, 5, 73, 367$
740	$371 + 2\sqrt{34410}$	$369 + 4\sqrt{8510}$	$16\sqrt{2139}$	$2, 3, 5, 23, 31, 37$
750	$376 + 5\sqrt{5655}$	$374 + 5\sqrt{5595}$	$2\sqrt{140621}$	$2, 3, 5, 13, 29, 373$
760	$381 + 2\sqrt{36290}$	$379 + 6\sqrt{3990}$	$12\sqrt{4011}$	$2, 3, 5, 7, 19, 191$
800	$401 + 20\sqrt{402}$	$399 + 20\sqrt{398}$	$4\sqrt{39999}$	$2, 3, 67, 199$
1000	$501 + 10\sqrt{2510}$	$499 + 10\sqrt{2490}$	$4\sqrt{62499}$	$2, 3, 5, 83, 251$

TABLE 3. The six instances of Theorem 8.2.

8.2. **What changes at two points.** The machine is that of Section 2 with z the root above: the same poles, atoms and functional, now with $A, B\colon \mathbb{Q}[t] \to \mathcal{K}$ and $\Delta_h \in \mathcal{K}[X]$. We write $y = z^2$; then $1 - y = -uz$. Since u is an algebraic integer with $u\,l(u) = 4$, u divides 4, and z, w are units of $\mathcal{K}$ (roots of a monic polynomial with constant term 1).

Reflection. Lemma 2.2 holds with $\sigma = s$, the generator of $\mathrm{Gal}(\mathcal{K}/\mathcal{F})$: its proof used only $\sigma(z) = -1/z$ and that the coefficients are rational functions of z with rational coefficients. So the odd-index entries lie in $e\,\mathcal{F}[X]$ with $e := z + 1/z$, $s(e) = -e$, $e^2 = u^2 + 4 = -nu \in \mathcal{F}$, and the rescaled matrix $E_{i+j}e^{[i \text{ odd}]-[j \text{ odd}]}$ shows $\Delta_h \in \mathcal{F}[X]$. The reflection removes the conjugates $-1/z$ and $-1/w$, where the series diverge.

The second place. The remaining conjugate, l, is not removed but convergent: the coefficients of A and B are rational functions of z, so $l(\Delta_h)(T_w) = \det(\Lambda_w(t^{i+j}))$, the Hankel determinant of the same machine at the weight w (the series converge since $0 < w < 1$). The two embeddings of $\mathcal{F}$ thus see the z-machine and the w-machine, and the endgame uses the norm $\mathcal{F} \to \mathbb{Q}$.

Odd primes. At every prime of $\mathcal{K}$ above an odd p, z, y and $1 - y = -uz$ are units. These are the only properties of z that Lemmas 3.1–3.5 and Theorem 3.7 use, so they hold verbatim, with the valuation normalised by $v(p) = 1$; there are no special odd primes, including the ramified ones and those dividing $n^2 - 16$. The bound is the same at every prime $\mathfrak{p}$ of $\mathcal{F}$ above p.

The prime 2. Here Lemma 4.1 fails: at $n = 400$, $h = 48$, the exact 2-adic content is 388.5, above $h\lfloor \log_2 h \rfloor + 3h = 384$. It is replaced by the following weaker bound.

Lemma 8.3 (Lemma 10″)**.** *Let n be even and $\mathfrak{P}$ a prime of $\mathcal{K}$ above 2, with $v = v_{\mathfrak{P}}$ normalised by $v(2) = 1$. Then $-v(\Delta_h) \le (h-2)^2/4 + 3h$.*

Proof. Since n is even, $u/2$ is a root of $X^2 + (n/2)X + 1$, hence a unit; so $\nu := v(1-y) = v(u) = 1$, instead of 2. Follow the proof of Lemma 4.1, which up to the valuations does not involve z. In (L), $y^b/(1-y)^{b+1}$ now has valuation $-(b+1)$, so $v(G_{ij}) \ge 1 - \pi + \min_{b<\pi}(v_2(b!) - 1) = -\pi$. In (U), with $v(\eta) = 1$, the first part has valuation $2 + s_2(N_0)$ and the second part valuation $\ge 1 + s_2(N_0)$, so $v(G_{ij}) \ge 2$ for $N_0 \ge 0$; the bound no longer grows with N_0. Hence $v(G_{ij}) \ge -f(i+j)$ with $f(s) := (s-K)_+$, a convex function, and $v(\tilde{\Delta}_h) \ge -\max_\sigma \sum_i f(i + \sigma(i))$. If $i < i'$ and $\sigma(i) > \sigma(i')$, put $c = i + \sigma(i)$, $d = i' + \sigma(i')$, $a = i + \sigma(i')$, $b = i' + \sigma(i)$; then $a + b = c + d$ and $a < c, d < b$, so convexity gives $f(c) + f(d) \le f(a) + f(b)$, and removing inversions does not decrease the sum. So the identity maximises, and $\sum_{i<h} f(2i) = 1 + 3 + \cdots + (h-3) = (h-2)^2/4$. Finally $-v(\Delta_h) = 3h - v(\tilde{\Delta}_h)$. □

So the prime 2 costs $\frac{1}{4}\log 2 = 0.1733\ldots$ per h^2 instead of 0. The mechanism that made it free in Lemma 4.1 is not available anywhere in the family:

Proposition 8.4. *For every n there is a prime $\mathfrak{P}$ of $\mathcal{K}$ above 2 with $v_{\mathfrak{P}}(1-y) \le 1$ (normalised by $v_{\mathfrak{P}}(2) = 1$). In particular $v_2(1-y) = 2$, which holds at $\sqrt{5} - 2$ and makes the prime 2 free there, cannot hold at every prime above 2.*

Proof. z is a unit, so $v_{\mathfrak{P}}(1-y) = v_{\mathfrak{P}}(u)$. From $u\,l(u) = 4$ we get $v_{\mathfrak{P}}(u) + v_{\mathfrak{P}}(l(u)) = 2$, and $v_{\mathfrak{P}}(l(u)) = v_{l^{-1}(\mathfrak{P})}(u)$. If $v_{\mathfrak{P}}(u) \ge 2$ for every $\mathfrak{P} \mid 2$, then $v_{\mathfrak{P}}(l(u)) \le 0$ for every $\mathfrak{P}$, so $v_{l^{-1}(\mathfrak{P})}(u) \le 0$, a contradiction. □

For even n, $\nu = 1$ at every prime above 2; for odd n, $\nu \in \{0, 2\}$, and the prime with $\nu = 0$ is worse, so odd n are not used. Proposition 8.4 does not show that the loss $\frac{1}{4}\log 2$ is necessary: exact profiles suggest that the true 2-adic cost is about $\frac{1}{8}\log 2$ per h^2, but we have not proved this.

The analytic part. Theorem 5.1 holds at a general weight $q \in \{z, w\}$, with $L := \log(1/q)$, the field $v_q(x) := -Lx - \Psi(x)$, and three changes. (a) v_q is strictly decreasing on $(-\tanh L, \infty)$, which contains $[-1/h, \infty)$ if and only if $h > \coth L$; this is the range of h ($\coth L_z = 364.9986\ldots$ at $n = 730$, and $\coth L_w < 1.0001$). (b) The dual density is only assumed ≥ 0, of mass 1, with $\operatorname{supp}\mu \subset [0, X_e]$ for some $X_e \ge 4$ (at the z-place the support is long, $X_e \approx 1300$), and $\Phi_q(\mu, \lambda) := -E(\mu) + \lambda + \int_0^{X_e} (F_\mu - \lambda)_+\,dx$ with $F_\mu = 2U^\mu + v_q$. (c) The tail condition becomes: with $B(x) := 2\log x - \log(x-1) - Lx$, both $\beta := \lambda - B(X_e)$ and $\gamma := L - (X_e - 2)/(X_e(X_e - 1))$ are positive. With $M_0 := 2\log(X_e + 1) + 1 + L - \lambda$ and $\kappa := 2/(1 - e^{-2\gamma})$, the proof of Theorem 5.1 gives, for every even $h > \coth L$,

$$\log\bigl|\det(\Lambda_q(t^{i+j}))_{i,j<h}\bigr| \le h^2\Phi_q(\mu, \lambda) + E_q(h),$$

$$E_q(h) := 2h\log(4h) + h\log 4 + (hX_e/2+2)\log 3 \\ + h\,c_1(h) + 2.362h - h\log h + h\max(M_0,0) + \kappa.$$

The pattern count below $X_e h$ replaces $(2h+1)\log 3$, the cell $[-1/h, 0]$ costs $h\max(M_0,0)$ (here $U^\mu \le \log(X_e+1)$ and $v_q \le L+1$ there), and the far nodes cost κ. Since $E_q(h) = O(h\log h)$, the lim sup is again Φ_q.

Nonvanishing. Lemma 6.1 holds over $\mathcal{F}$ for $h = 3\ell - 1$ with $\ell \ge 7$ prime and unramified in $\mathcal{K}$. Now $\zeta := z^\ell \bmod \mathfrak{l}$ is $\mathrm{Frob}_\mathfrak{l}(z)$, one of four values:

$\mathrm{Frob}_\mathfrak{l}$	1	s	l	sl
ζ	z	$-1/z$	w	$-1/w$
$1-\zeta^2$	$-uz$	u/z	$-(4/u)w$	$(4/u)/w$
$1-\zeta^{-2}$	u/z	$-uz$	$(4/u)/w$	$-(4/u)w$

Every entry is an $\mathfrak{l}$-unit because $u \mid 4$ and ℓ is odd, and steps 1–5 of Lemma 6.1 use only these units and $1 - z^{-2} = u/z$. Since ℓ is unramified, $v_\mathfrak{l}$ restricts to $v_\mathfrak{L}$ on $\mathcal{F}$ for $\mathfrak{L} = \mathfrak{l} \cap \mathcal{F}$, and step 6 gives $\Delta_h(\alpha) \ne 0$ for every $\alpha \in \mathcal{F}$ with $v_\mathfrak{L}(\alpha) \ge -1$, whether ℓ is split or inert in $\mathcal{F}$.

Lemma 8.5 (Lemma E′, the norm endgame)**.** *Let $n \ge 6$ be even, and suppose that for each $q \in \{z, w\}$ there is a pair (μ_q, λ_q) as in (b) above ($\mu_q \ge 0$, mass 1, $\operatorname{supp}\mu_q \subset [0, X_e]$ with $X_e \ge 4$; the cap $\mu_q \le 1$ is not required) satisfying the tail condition (c), with*

$$m(n) := \tfrac{1}{2}\big(\Phi_z(\mu_z,\lambda_z) + \Phi_w(\mu_w,\lambda_w)\big) + \tfrac{4}{3} + \tfrac{1}{4}\log 2 < 0.$$

Then M1(n) *holds.*

Proof. Suppose $\alpha \in \mathcal{F}$ has $T_z = \alpha$ and $T_w = l(\alpha)$, and fix $b \ge 1$ with $b\alpha \in \mathcal{O}_\mathcal{F}$. Let $\mathcal{H} := \{3\ell - 1 : \ell \ge 7 \text{ prime, unramified in } \mathcal{K},\ \ell \nmid b,\ 3\ell - 1 > \coth L_z\}$, an infinite set; note $\coth L_w < \coth L_z$. For $h \in \mathcal{H}$, Lemma P′ over $\mathcal{F}$ gives $\Delta_h(\alpha) \ne 0$. For a prime $\mathfrak{p}$ of $\mathcal{F}$ put $n_\mathfrak{p} := -v_\mathfrak{p}(\Delta_h)$, which may be a half-integer at ramified primes, and $n_p := \max(0, \max_{\mathfrak{p}\mid p}\lceil n_\mathfrak{p}\rceil)$, $c_h := \prod_p p^{n_p}$. Then $c_h\Delta_h \in \mathcal{O}_\mathcal{F}[X]$, and by the odd-prime lemmas and Lemma 8.3,

$$\log c_h \le \sum_{3\le p<K} B_p \log p + \big((h-2)^2/4 + 3h\big)\log 2 + \theta(K) + \log 2,$$

where B_p are the bounds used in Theorem 3.7; the term $\theta(K) = O(h)$ pays for all ceilings. So $\limsup_{h\in\mathcal{H}} h^{-2}\log c_h \le \frac{4}{3} + \frac{1}{4}\log 2$. The algebraic integer $\beta := b^h c_h \Delta_h(\alpha)$ is nonzero. Its two real embeddings are $b^h c_h \Delta_h(T_z) = b^h c_h \det(\Lambda_z(t^{i+j}))$ and $b^h c_h\, l(\Delta_h)(T_w) = b^h c_h \det(\Lambda_w(t^{i+j}))$, so $N_{\mathcal{F}/\mathbb{Q}}(\beta) = b^{2h} c_h^2 \det(\Lambda_z(t^{i+j}))\det(\Lambda_w(t^{i+j}))$ is a nonzero integer. Taking logarithms, dividing by $2h^2$, and applying the analytic bound at both places, $0 \le \frac{4}{3} + \frac{1}{4}\log 2 + \frac{1}{2}(\Phi_z + \Phi_w) = m(n) < 0$, a contradiction. □

8.3. **The certificates.** For each $n \in \mathcal{N}$ and $q \in \{z, w\}$ the certificate is a piecewise constant $\mu_q \ge 0$ on a grid of $[0, X_e]$ with mass exactly 1, a rational λ_q, a rigorous Arb upper bound for $\Phi_q(\mu_q, \lambda_q)$, and an Arb check

of the tail condition with $X_e \geq 4$; the logarithm $L = \log(1/q)$ is an Arb ball computed from the quartic, so every bound encloses the value at the true L. Two independent evaluators were used, and where both certified the same (n, q) the smaller bound is taken. Table 4 lists the results; $\frac{4}{3} + \frac{1}{4}\log 2 = 1.50662012847331966\ldots$, and every $m(n)$ is computed in Arb from the printed balls and lies entirely below 0.

n	$\Phi_z \leq$	$\Phi_w \leq$	$m(n) \leq$
730	0.97551590832772	−3.98960878950	−0.000426312
740	0.98154355971108	−3.99641267680	−0.000814430
750	0.98568546476333	−4.00312517276	−0.002099725
760	0.98610919549783	−4.00974867868	−0.005199613
800	1.00091072617	−4.03494034059	−0.010394678
1000	1.05765433525	−4.14651125316	−0.037808330

TABLE 4. Certified bounds for Theorem 8.2 (upper endpoints of Arb balls, rounded up). At the z-place $X_e = 1300$ for $n \leq 760$, and $X_e \approx 1202$ and 1503 for $n = 800$, 1000; at the w-place $X_e \approx 4$.

The constant Φ_z is positive because $z \approx 1 - 2/n$ gives the z-machine almost no decay; the w-place, with $w \approx 1/n$, pays for it. The side constants are comfortable: at $n = 730$, for example, $\beta = 0.5719$, $\gamma = 0.001971$, $M_0 \leq 11.17$, $\kappa \leq 508.4$ at the z-place, and $\beta = 16.41$, $\gamma = 6.426$ at the w-place. A third evaluator, written independently of both, confirms $m(n) < 0$ for all six n and gives $m(730) \leq -0.000452$. As a validation, the three codes applied at Theorem 1.1's weight $\sqrt{5} - 2$ give -1.35926599, -1.35939764 and -1.35939884, consistent with Theorem 5.3 and the primal value -1.3594016 of Remark 5.4.

8.4. **Scope.** We state the limits of Theorem 8.2 explicitly.

- It is a disjunction. We do not know which of the two statements holds, and neither is known by other means: $T_n \notin \mathbb{Q}$ is a statement of strong-four-exponentials type, and we know of no result on $T_z - T_w$.
- For very large n a general criterion may apply: our evaluation of [4, Thm. 1] with the shift $\frac{1}{2}$ at w^2, at the real place where w is small, suggests that its condition holds once $n \gtrsim 7.6 \cdot 10^{13}$, which would give $\chi_2(w) \notin \mathcal{K}$ and hence M1(n) for those even n. This evaluation is a sketch that has not been checked in detail or by a referee. For $730 \leq n \leq 1000$ the same quantity lies between -12.7 and -12.5, so Theorem 8.2 is not implied.
- It is proved for the six listed values of n only, each with its own certificates. Even n between them are not covered, and odd n are not covered at all: Lemma 8.3 needs n even, and for odd n one prime above 2 has $v(1 - y) = 0$.

- For all even $n \geq 730$ the statement is supported only numerically. A proof would need interval certificates in n (the dual bound is monotone in L, but this needs to be written) and an argument as $n \to \infty$; neither exists.
- The method cannot reach $n = 720$: there the primal value of the equilibrium problem gives $m(720) \geq +0.00098$ (computed numerically, with a density admissible up to rounding of order 10^{-12}), so no dual certificate can succeed. Numerically the threshold of the method lies in $(720, 730)$.
- The theorem is not formalised in Lean. Its proof was checked by two AI referees and has not been read by a human.

9. The formalisation

9.1. **What is formalised.** The Lean project [16] proves Theorem 1.1 and the identity (1.1) in Lean 4 [6] (version v4.35.0) with Mathlib [28] (version v4.35.0 [29]). It has 62 modules under `Chi2U/`, with about 16,700 lines, of which about 7,800 are ported verbatim from the formalisation of [14] [15]: the prime number theorem, the interval arithmetic, the generic analytic files, the criterion and the definition of χ_2. All declarations are in the namespace `Chi2U`. The project is available at `https://github.com/telaviv/pi2-log2-golden-irrationality` [16], together with this paper and the scripts of Section 10. Theorem 8.2 is not formalised.

9.2. **The statement.** The definitions that the reader has to check are in `Defs.lean`, `Statement.lean` and `Golden/Corollary.lean`:

```
noncomputable def chi2Term (z : ℝ) (n : ℕ) : ℝ :=
    z ^ (2 * n + 1) / (2 * (n : ℝ) + 1) ^ 2
noncomputable def chi2 (z : ℝ) : ℝ := ∑' n : ℕ, chi2Term
z n
def TheoremU : Prop := Irrational (chi2 (Real.sqrt 5 - 2))
theorem chi2_sqrt5_sub_two_irrational : TheoremU
theorem chi2_sqrt5_sub_two_eq :
    chi2 (Real.sqrt 5 - 2) = Real.pi ^ 2 / 24 - 3 / 4 *
Real.log goldenRatio ^ 2
theorem golden_irrational' :
    Irrational (Real.pi ^ 2 - 18 * Real.log goldenRatio ^ 2)
```

The first theorem is in `Main.lean`, the second in `Golden/ClosedForm.lean`, the third in `Golden/Corollary.lean`. The definition of χ_2 is identical to that of [15], and `Defs.lean` proves summability for $|z| \leq 1$, so `chi2 (Real.sqrt 5 - 2)` is the genuine sum of the series. `goldenRatio` is Mathlib's $(1+\sqrt{5})/2$ and `Irrational` is Mathlib's predicate $x \notin \mathbb{Q}$. The identity (1.1) is proved from Landen's identity for χ_2, Euler's reflection formula and Landen's identity for Li_2 (`Golden/Landen.lean`, `Golden/Dilog.lean`), so the golden-ratio form is not a separate assumption.

The command `lake env lean Axioms.lean` prints

```
'Chi2U.chi2_sqrt5_sub_two_irrational' depends on axioms:
    [propext, Classical.choice, Quot.sound]
'Chi2U.golden_irrational'' depends on axioms:
    [propext, Classical.choice, Quot.sound]
```

and the same list for `chi2_sqrt5_sub_two_eq` and for each of the main ingredients listed in `Axioms.lean`. These are Lean's three standard axioms. The development contains no `sorry`, no added axiom, no `native_decide`, no `implemented_by` and no `@[extern]`, so no step relies on compiled code.

9.3. **Correspondence with the paper.** Table 5 lists the main formal counterparts. Paths are relative to `Chi2U/`.

paper	Lean declaration (file)
$\mathbb{K}$, z, σ	`Q5` = `QuadraticAlgebra` $\mathbb{Q}$ 5 0, `zK`, `sig`, `emb` (`Field`)
Lemma 2.1, (2.1)	`lemma1`, `aeval_chi2_Delta` (`PoleExpansion`); `Lmom_eq_mahler` (`Mahler`)
Lemma 2.2, Cor. 2.3	`eval`$_2$`_Delta`, `aeval_Delta`, `Delta_map` (`LemmaR`)
Prop. 2.4	`irrational_of_poly_approx` (`Criterion`); `MachineBoundsU`, `irrational_of_boundsU` (`MainCond`)
Lemma 3.3	`AK_eq` (`Arith/ClassCount`)
Lemma 3.4	`lemmaB` (`Arith/LemmaB`)
Lemmas 3.2, 3.5	`lemma9` (`Arith/OddPrimes`, `Arith/Lemma9Levels`); `Arith/Lemma8One`; `padicNorm_Delta_odd` (`Arith/OddExp`)
Thm. 3.7	`theoremA` (`Arith/PrimeSums`); `denAll_int`, `log_denAll_le` (`Arith/Assembly`)
Lemma 4.1	`lemma10_L`, `lemma10_U`, `padicNorm_Delta_two` (`Arith/TwoAdic`)
Thm. 5.1 (other route)	`log_abs_Dpoly_ge`, `DualCertU`, `DualCertU.analytic` (`Analytic/TheoremBU`)
Thm. 5.3 (coarser)	`exists_certU` (`Cert/CertU`); data `Cert/DataU`
Lemma 6.1	`lemmaP`, `nonvan_frequently` (`NonVan/LemmaP`)
Section 7	`boundsU`, `chi2_sqrt5_sub_two_irrational` (`Main`)
(1.1)	`chi2_landen`, `Li2_reflection`, `Li2_landen` (`Golden/Landen`); `chi2_sqrt5_sub_two_eq` (`Golden/ClosedForm`)

TABLE 5. Formal counterparts of the main steps.

9.4. **Differences from the written proof.** The formal proof establishes what the endgame needs, and in several places it takes a route that was shorter in Lean.

- *The machine.* Δ_h is *defined* as the determinant of the rational coordinates of the matrix $\tilde{E}$ of Corollary 2.3, so $\Delta_h \in \mathbb{Q}[X]$ holds by definition; Lemma 2.2 and Corollary 2.3 then identify it with the Hankel determinant over $\mathbb{K}$ and its value at T with $\det(\Lambda(t^{i+j}))$.

- *Theorem B'.* The formal proof does not follow Section 5. It follows the lattice-energy route of the formal proof of Theorem B in [14]: the Andreief identity for the node expansion, the bound $h^h(1+2/h)^{h^2}$ for determinants with derivative rows via averages over roots of unity (instead of step 3), and a discrete Pólya-type lattice energy inequality comparing the patterns, which live on the odd nodes, with a reference sequence on all of $\mathbb{N}$ (instead of steps 5–6). The ingredient not present in [15] is the two-sided Jensen bound of step 4 (`log_abs_Dpoly_ge`), which gives the node weight (5.3). This route evaluates the dual at the nodes pointwise, so it never meets the cell $[-1/h, 0]$ and does not use the monotonicity (5.1) (which is nevertheless proved, as `vU_strictAntiOn`, on $[-\frac{1}{2}, \infty)$). It is stated asymptotically, like Theorem B of [14]: the explicit $E_{\mathrm{an}}(h)$ is not formalised.
- *The certificate.* The formal certificate (`DualCertU`) consists of a step density with 50 cells of $[0, 2]$ (49 nonzero), grid width $1/L$ with $L = 200$ on $[0, X]$ with $X = 4$, rational cell bounds S_k for $2U^\mu + v$, and the tail conditions; its exponent is $-E(\mu) + \lambda + \frac{1}{L}\sum_k (S_k - \lambda)_+$. The cell bounds use chords and tangents of each term, as in [14], which keeps the cancellation between neighbouring cells without the near/far split of Section 5.3. The certified value is $\Phi_{\mathbb{Q}} = -54155169767979/40000000000000 \approx -1.3538792$, coarser than Theorem 5.3 but enough: the assembly (`boundsU`) uses $\beta = -27/20$, so the formal margin is $\frac{4}{3} - \frac{27}{20} = -\frac{1}{60}$.
- *Lemma $8'$.* Only the single-level range $K/3 \le p$ used by Theorem 3.7 is formalised, by a crude row bound on the Lagrange Gram matrix and Lemma 3.4 instead of the atom expansion.
- *Lemma P'.* Congruences are taken in the residue field $\mathbb{Z}_{(\ell)}[\varphi]/\mathfrak{l}$ with $\zeta := z^\ell$ itself; no identification of ζ by Frobenius is needed, only $1 - \bar{z}^{2\ell} = (4\bar{z})^\ell$. The progression sums are handled by an exact shift identity instead of the Newton form.
- *The endgame* uses only even h, through $P_h := \Delta_{2\lfloor h/2 \rfloor}$ in the criterion of [15].

9.5. **How the certificate is checked.** The certificate data (`Cert/DataU.lean`) were produced by the script `scripts/gen_cert_u.py` (run as `python scripts/gen_cert_u.py 50 8 4 Chi2U/Cert/DataU.lean`). The script is not trusted: every check is a decidable statement about the rational data, proved by `decide +kernel` in `Cert/CertU.lean` in 16 chunks of grid cells, together with a table of 2004 logarithm enclosures, the energy bound and the tail. The dyadic enclosure of $\log 2$ used by the table is itself checked by `decide +kernel`. A wrong value would make the build fail but could not make a false statement provable.

9.6. **Cost.** The kernel check of the certificate takes about 6.5 minutes. A build of a fresh clone of the repository, with Mathlib's prebuilt cache and at most four parallel Lean processes, took 9 minutes 35 seconds (9110 jobs, no errors) on a Windows 11 desktop with 32 logical processors and 31.7 GB of RAM, and `Axioms.lean` then reported only the three standard axioms.

9.7. **What has to be trusted.** Beyond the Lean kernel and the three axioms, the reader has to trust that the definitions displayed above express Theorem 1.1, and the Mathlib definitions they use (real numbers, `tsum`, `Real.sqrt`, `Real.log`, `Real.pi`, `goldenRatio`, `Irrational`). Nothing else, in particular no numerical data and no generating script, needs to be trusted. To avoid trusting the elaborator's handling of the environment, every declaration of the 62 project modules was replayed through the kernel with `leanchecker` (shipped with the Lean toolchain), each module against the compiled environment of its imports, on 7 November 2026; all modules passed, in about 42 minutes in total. Mathlib itself was not replayed from scratch (`--fresh`). An AI referee checked the fidelity of the statements (the definitions above, traced to Mathlib, and the absence of hypotheses, custom axioms and compiler trust) and found them faithful. Apart from the check by the Lean kernel, all verification of the formalisation so far was carried out by AI systems; no human expert has yet reviewed the Lean code.

10. Computations and reproducibility

10.1. **Software.** The computations use python-flint 0.9.0 [27]: exact arithmetic over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$ for every statement about valuations, and Arb ball arithmetic [11] for every real number that feeds a claim. Hankel determinants are evaluated at T in ball arithmetic with working precision about $3.6h^2 + 8000$ bits. The formalisation uses Lean v4.35.0 and Mathlib v4.35.0.

10.2. **What each computation is used for.**

- *Proof inputs.* The certificate of Theorem 5.3 (`cert_U.py`, run with arguments `1600 32 2 4`, 128 bits, about one minute); the certificates of Theorem 8.2 (`dual_ref.py`, and `dualcert_arb2.py` with its library `dualcert.py`), the margins and the explicit numbers of Section 8 (`facts_M1.py`, Arb at 3000 bits). These are rigorous: floating point is used only to choose the density and λ, and any choice gives a valid bound.
- *The effective threshold* of Remark 7.1 (`h0_U.py`, `h0_scan.py`): exact sums, using the constants of Theorem 5.1.
- *Checks of the lemmas* (not used by the proof). The machine and the matrix $\tilde{E}$ (`umach_core.py`); the bordered identity at $h = 2, \dots, 12$ and $X \in \{0, 1, -3/7\}$ (`bordered_check.py`); the row claim (`rowclaim_check.py`); Lemmas 3.2, 3.5 on exact profiles up to $h = 256$ (`law_check.py`, `u_profile.py` with `fastcontent.py`); Lemma 3.3 by brute force for every odd $d < K + 40$ and

194 values of K (`law_check.py`); Lemma 4.1 entrywise for $h \le 64$ (`lemma10_check.py`); Lemma 6.1 (`blocks_check.py`, `lemmaP_check.py`). An AI referee reimplemented the machine without sharing code and reproduced the numbers it checked; the certificate reproduced bit for bit at $N = 100, 200, 1600$.

These scripts are in the folder `scripts/paper` of the repository [16] (those for Section 8 in `scripts/paper/m1`), with a README that gives the command line for each of them and the small result files quoted here. Not included are the referees' own check scripts (apart from `dual_ref.py`) and the earlier script that produced the exact profiles for $h \le 128$ in Table 6; those profiles are included as data, and `u_profile.py` computes the same quantities (checked at $h = 16$).

10.3. **Numerical evidence.** Table 6 shows exact scores. The arithmetic column is $h^{-2}\sum_p n_p \log p$ for the exact contents, so the score is the sum of the two previous columns. The values approach the limits -1.3592 (or better) and $\frac{4}{3}$ slowly, as $O(\log h/h)$.

h	score $h^{-2}\log\lvert F_h(T)\rvert$	analytic	arithmetic	n_2
16	-0.2930	-1.4379	1.1449	54
32	-0.1734	-1.4195	1.2462	142
64	-0.1147	-1.4015	1.2869	350
96	-0.0951	-1.3930	1.2980	576
128	-0.0764	-1.3876	1.3112	830
192	-0.0659	-1.3815	1.3156	1344
256	-0.0596	-1.3781	1.3185	1918

TABLE 6. Exact scores for Theorem 1.1. At $h = 192$ and 256 the contents are proved lower bounds for each n_p (fast content computation), which can only make the score look worse.

10.4. **Controls.** No false instance of the method is available, since no unit with rational χ_2 is known. As substitutes: the proved finite-h bound of Remark 7.1 lies above the exact score at every tested h, and every per-prime inequality holds on the exact profiles, often with equality; for a unit of norm $+1$ the reflection breaks; with poles on one side only the conjugate place costs $+0.50$ per h^2 at $h = 16$; and at the points $\sqrt{2} - 1$ and φ^{-1} the corresponding scores are positive (primal lower bounds $+0.3286$ and $+0.6572$ for the limiting exponent), so the method does not conclude there. For Theorem 8.2, the method does not conclude at $n = 6$ ($m(6) \approx +0.477$), $n = 400$ ($m(400) \ge +0.070$) or $n = 720$ (numerical values).

Acknowledgements and disclosure

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